Aerospace Science and Technology 169 (2026) 111491

Contents lists available at ScienceDirect

Aerospace Science and Technology

journal homepage: www.elsevier.com/locate/aescte

Model predictive control for aircraft engine thermal management under nonlinear heat transfer and time delay

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ARTICLE INFO

Edited by: Fang Xiande

Keywords:
Aircraft engine
Thermal management system
Model predictive control
Heat current method
Time-delay effect

ABSTRACT

Growing thermal loads and frequent engine shifts in advanced aircraft pose critical thermal management challenges. Fuel thermal management systems (FTMS) enable heat dissipation from distributed sources but exhibit nonlinearity and strong coupling that challenge conventional control methods. Although model predictive control (MPC) has been applied, its dependence on linearized models limits effectiveness across wide operating conditions. This paper applies the heat current method to derive a linearized equivalent model from the original nonlinear heat exchange constraints, explicitly incorporating time-delay effects. This model enables the design of a constrained MPC strategy, which seamlessly integrates real-world system characteristics to achieve robust and fully automated thermal management across wide operating conditions. Experimental validation on a full-scale three-medium (fuel/water/air) ground-test rig with an intermediate circuit demonstrates the effectiveness of the proposed strategy under realistic flight conditions. The system maintains air outlet temperature within strict operational limits while ensuring complete prevention of water boiling, achieving a maximum temperature deviation of 3.43 K and a time-averaged deviation of 0.96 K. These quantitative results confirm the controller's robust performance in managing complex thermal dynamics under strong time-varying operating conditions.

1. Introduction

The rapid development of hypersonic vehicles (Ma $\geq$ 5) and electric propulsion systems has led to a sharp increase in onboard heat loads, while simultaneously reducing opportunities for heat rejection, thereby subjecting thermal management systems to increasingly extreme conditions [1]. On the heat source side, the incorporation of high-power electronics, phased array radars, and directed energy weapons has significantly elevated cooling demands [2]. On the heat rejection side, stealth-driven designs restrict external air inlets, thereby limiting available cooling airflow. The adoption of lightweight composite materials, exhibiting reduced thermal conductivity, impedes efficient heat transfer [3]. Meanwhile, supersonic and hypersonic flight conditions increase stagnation temperatures, further diminishing heat sink capacity [4]. Furthermore, frequent operational shifts, such as during takeoff, climb, and acceleration, introduce sharp fluctuations in heat load, giving rise to complex control challenges including pronounced nonlinearities, strong cross-coupling, and significant time delays. These factors collectively make the design and control of thermal management systems a critical challenge in modern aviation and high-performance vehicle development [5].

Recent studies have explored various aspects of thermal management systems, including high-temperature component cooling [6], waste heat recovery and energy integration [7] and intelligent thermal management [8]. As the core power unit, the engine not only generates substantial heat but also assists in dissipating thermal energy from other systems. Fuel circulating within the engine can act as a coolant, enabling the rejection of excess heat. Consequently, fuel thermal management systems (FTMS) have become a major research focus, with the primary objective of ensuring engine safety and efficiency under extreme thermal conditions while maximizing energy utilization.

From the perspective of classical control theory, several approaches have been applied to FTMS. For instance, Zhou et al. [9] introduced a fuzzy self-tuning PID strategy for heat exchangers and electronic components, while Yu et al. [10] adopted conventional PID to independently

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https://doi.org/10.1016/j.ast.2025.111491
Received 18 October 2025; Received in revised form 25 November 2025; Accepted 7 December 2025
Available online 9 December 2025
1270-9638/

Nomenclature

A	Heat transfer area,
c	Opening degree, %
c_p	Specific heat capacity, J $kg^{-1}\cdot K^{-1}$
e	Derivation
G	Heat capacity rate, $W\cdot K^{-1}$
m	Quality, kg
$\dot{m}$	Mass flow rate, kg s^{-1}
K	Heat transfer heat coefficient, $W\cdot K^{-1}$ m^{-2}
Q	Heat transfer rate, W
r	Reference signal
R	Thermal resistance, K W^{-1}
t	Time, s
T	Temperature, K
u	Input variable
w	Frequency, Hz
x	State variable
y	Output variable
ε	Thermal potential, K
τ	Delay time, s

Subscripts

c	Cold side
cin/cout	cold side inlet/outlet
d	delay
fp	Fluid pipeline
h	Hot side
hin/hout	hot side inlet/outlet
H1-H3	Heat exchangers 1–3
mix	Mixing process
T, D, M	Tank, distributor, merger
0,1,2	Main loop, branch 1, branch 2

Abbreviation

AHE	Air-water Heat Exchanger
FTMS	Fuel Thermal Management System
FHE	Fuel-water Heat Exchanger
MPC	Model Predictive Control
TMS	Thermal Management System

regulate multiple pumps and valves, determining parameters through iterative testing. However, these strategies predominantly rely on **localized control designs** and significant empirical tuning. They fundamentally struggle to account for **system-wide coupling effects** and lack a predictive capability to handle time delays. Consequently, optimizing parameters for such decentralized controllers remains challenging, and their performance under widely varying operating conditions is often uncertain and unreliable.

Some studies employ global models and search-based methods for control. Yang et al. [11] adopted the logarithmic mean temperature difference model of the heat exchanger and ignored the fluid flow delay, and developed a global controller based on a Particle Swarm Optimization algorithm with a Minimum Variation Distance to suppress oscillations caused by multiple optimal solutions. While effective in identifying optimal setpoints, such methods can still yield multiple parameters sets with identical control effects under certain conditions. This may necessitate frequent adjustments of valves and pumps in practice, thereby increasing operational complexity. Simultaneously, the computational cost of this method increases dramatically when considering complex dynamics like fluid flow delays, rendering it impractical for real-time online control applications.

As a prominent strategy for multi-variable constrained systems, Model Predictive Control (MPC) is a natural candidate for FTMS [12], However, its application is fundamentally constrained by the conflict between its linear model requirement and the nonlinear, delayed dynamics inherent to FTMS. This conflict forces a common compromise in the literature: the use of simplified linear models that fail to fully capture the system's core physics. These simplifications primarily manifest in two aspects: the **neglect of essential dynamics** and the **oversimplification of heat exchange**. The resulting limitations in the existing literature are detailed as follows.

Arcos-Aviles et al. [13] implemented MPC in an integrated electro-thermal system using an energy balance model analogous to an Energy Hub. However, their model did not incorporate practical energy transport constraints, such as fluid flow delays. Deppen et al. [14] made an assumption of a constant temperature difference to derive linear heat transfer constraints for FTMS control. A critical drawback of this approach is that their model does not account for fluid flow delays, which can degrade control performance. Doman [15] simplified the system by neglecting the thermal capacitance of the fuel tank, holding the heat exchanger inlet temperature constant and thereby ignoring its dynamic impact on heat exchange efficiency. Koeln et al. [16] adopted a hierarchical MPC for aircraft electro-thermal systems and provided a linear heat exchanger equation based on the assumption of a uniform fluid temperature within the exchanger, which is a significant simplification of the actual temperature distribution.

A more prevalent simplification involves treating the complex heat transfer process as a predetermined boundary. Leister et al. [17] assumed that the heat transfer rate of the heat exchanger was a given boundary condition, thereby neglecting the impact of dynamic temperature and flow rate changes on the heat transfer process. Under the same assumption, Sigthorsson et al. [18] simplified the fluid flow delay as an equivalent heat capacity effect, using the outlet temperature as the characteristic state variable. This approximation, however, fails to represent the full spatiotemporal delay characteristics, potentially leading to inaccuracies in dynamic control performance. Although Oppenheimer et al. [5] made progress by explicitly modeling the fluid flow delay using a delay time and deriving a linear state-space model via state augmentation, this approach still fell short of fully capturing the inherent nonlinearities in the heat transfer process itself.

For clarity, the main features and simplifications of the reviewed literature are summarized in Table 1. Crucially, the collective neglect or oversimplification of these transport characteristics introduces systematic bias into the prediction model, thereby degrading the overall control performance and robustness across wide operating conditions.

Table 1
The main features of the reviewed literature related to MPC.

Literature	Year	Model Feature for heat exchange	Model Feature for fluid flow delay
Arcos-Aviles et al. [13]	2024	Energy hub model	×
Deppen et al. [14]	2015	Constant heat exchange temperature difference	×
Doman et al. [15]	2015	Constant inlet temperature	×
Koeln et al. [16]	2019	Fluid temperature is uniform	×
Leister et al. [17]	2022	Given heat transfer rate	×
Sigthorsson et al. [18]	2019	Given heat transfer rate	Approximate based on heat capacity effect
Oppenheimer et al. [5]	2019	Given heat transfer rate	Delay time

In summary, the preceding review underscores a critical gap: a pressing need for control-oriented modeling approaches that faithfully preserve essential system characteristics, particularly nonlinear heat transfer and spatiotemporal delay effects. The *heat current method* emerges as a promising foundation to address this gap. This methodology was pioneered by Chen et al. [19], who redefined the concept of thermal resistance and formulated a thermal Ohm's law, thereby establishing a linear constitutive relation between temperature difference and heat flow for modeling nonlinear heat exchange. The framework was subsequently generalized by introducing concepts such as thermal potential and heat source, which systematically characterize energy conservation, dynamic storage, and delay properties inherent in thermal processes [20]. As a result, the heat current method offers a unified approach that holistically captures the **nonlinear, dynamic, and delayed characteristics** critical to FTMS. This comprehensive capability makes it particularly suitable for integration with MPC. Nevertheless, methodologies for converting such high-fidelity thermal circuit models into control-oriented predictive models suitable for real-time MPC implementation remain underdeveloped, presenting a key challenge that this work aims to overcome.

This paper presents a constrained MPC strategy for aircraft engine FTMS, founded upon a **control-oriented linearized equivalent model** derived from the heat current method. The primary objective of this work is to develop a robust and automated control framework that effectively handles the nonlinearities and time-delay effects prevalent in FTMS, which are often oversimplified in existing works [5,13–18]. To achieve this, the proposed approach systematically addresses the key limitations identified in the literature: first, a high-fidelity system model is constructed using the heat current method, **explicitly incorporating nonlinear heat transfer constraints and time-delay effects** that are often oversimplified in existing works; second, this model is transformed into a state-space representation suitable for MPC, enabling the design of a constrained predictive controller with integrated feedback correction to enhance robustness; finally, the proposed strategy is **rigorously validated on a full-scale ground-test experimental system** under strong time-varying flight conditions, demonstrating its effectiveness in stabilizing outlet temperatures while preventing the intermediate medium from boiling.

2. Modeling

Fig. 1 illustrates the partial of a real aircraft engine FTMS with an intermediate circulation loop. Low-temperature fuel exits the fuel tank, passes through a heat exchanger to cool the high-temperature lubricating oil, then splits: one stream cools external circulating water via a heat exchanger, while the other flows through a fuel-water heat exchanger (FHE). After that, two streams converge into fuel supply header to enter the engine. In the intermediate circulation loop shown in the dashed box, high-temperature air transfers heat to water via two air-water heat exchangers (AHEs), which is then absorbed by low-temperature fuel from the FHE. Arrows indicate flow directions: black for water, orange for high-temperature air, blue for fuel, red for lube, and light blue for external circulating water. From a holistic perspective, the low-temperature fuel collects heat from high-temperature lube, external circulating water and air, while utilizing an intermediate circulation loop to reduce heat exchange difference and improve safety. This study focuses exclusively on the control for the **Intermediate Circulation Loop**, in direct response to a practical requirement,

Considering that the finite velocity of fluid flow in pipelines and heat exchangers causes delays in temperature response, in Fig. 1, τ represents the delay for each pipeline section, and the subscripts T, D, H1∼3, and M represent the water tank, distributor, heat exchanger, and merger, respectively. $\dot{m}$ is the mass flow rate, and the subscripts 0, 1, 2, h1, and h2 represent the main loop, branch 1, branch 2, high-temperature air 1, and high-temperature air 2, respectively. T stands for temperature, and the subscripts Ti, To, hin1, hin2, M, hout1, and hout2 represent the inlet and outlet temperatures of the water tank, the inlets of high-temperature air 1 and high-temperature air 2, the outlet at the merger, and the outlets of high-temperature air 1 and high-temperature air 2, respectively. Q represents the heat flow rate.

2.1. Modeling of components

The system comprises pipelines, heat exchangers, a water tank, and diverting/mixing processes. The following assumptions are made:

(1) Delay time is flow-rate-independent.

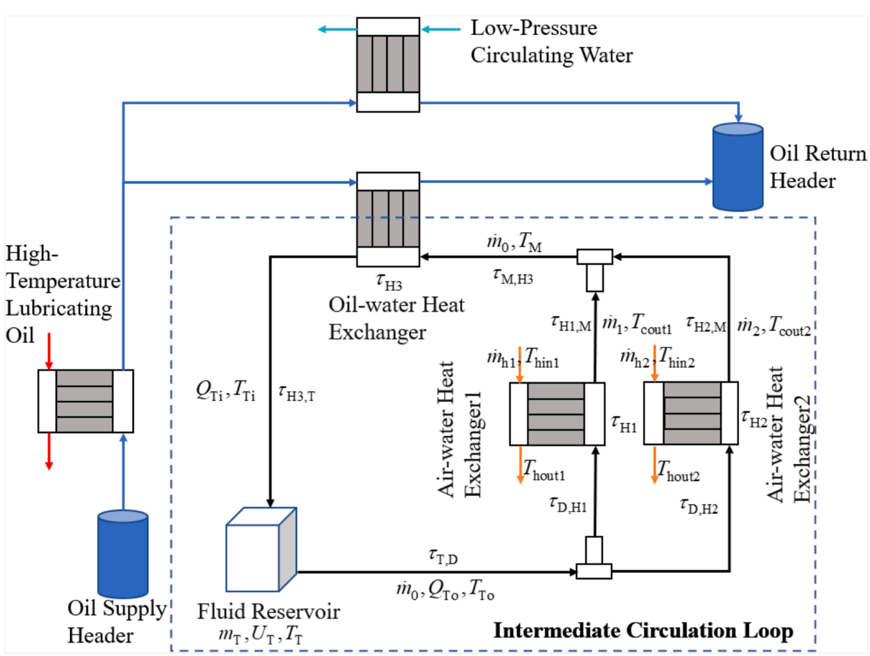


Fig. 1. The partial of a real aircraft engine FTMS with an intermediate circulation loop.

(2) Water tank temperature is uniform, and outlet temperature equals tank temperature.

Using the heat current method, the intermediate loop system is analyzed and modeled. Component models are as follows:

(1) Pipeline

The former research has given an entire dynamic heat current model of pipeline with the consideration of the complex coupling relationships among heat transfer, migration and storage [20]. In this study, we neglect the influence of heat loss to the environment and only study the time delay between inlet and outlet as

$$T_{\text{out}}(t) = T_{\text{in}}(t - \tau_{\text{fp}}), \tag{1}$$

where T_{out} and T_{in} denote the pipeline's outlet and inlet temperatures, respectively, t represents the current time, and τ_{fp} is the delay time, defined as the time taken for the working fluid to pass through the pipeline.

Therefore, Fig. 2 shows a simplified heat current model of a pipeline, utilizing a *delay thermal potential* $\varepsilon_{\text{d}}(t)$ to reflect the constraint of Eq. (1) with the expression as

$$\varepsilon_{\text{d}}(t) = T_{\text{in}}(t) - T_{\text{in}}(t - \tau_{\text{fp}}). \tag{2}$$

(2) Mixing and Splitting

A mixing process involves n tributaries, possessing mass flow rate $\dot{m}_i$ and temperature T_i ($i = 1\sim n$), converging into a main stream with temperature T_{mix}. The conservation of energy could be represented by

$$\left(\sum_{i=1}^{n} \dot{m}_i\right) c_p(T_{\text{mix}}) T_{\text{mix}} = \sum_{i=1}^{n} \dot{m}_i c_p(T_i) T_i, \tag{3}$$

where $c_p(T)$ is the specific heat capacity of fluid with temperature T. Assuming that $c_p(T)$ is constant during the fluid mixing process, Eq. (3) can be simplified as

$$T_{\text{mix}} = \sum_{i=1}^{n} \dot{m}_i T_i \Big/ \sum_{i=1}^{n} \dot{m}_i, \tag{4}$$

On this basis, Fig. 3 gives the heat current model of this mixing process, utilizing *mixing thermal potentials* $\varepsilon_{\text{mix},i}$ to reflect the constraint of Eq. (4) with the expression as

$$\varepsilon_{\text{mix},i} = \sum_{i=1}^{n} \dot{m}_i T_i \Big/ \sum_{i=1}^{n} \dot{m}_i - T_i (i = 1 \sim n). \tag{5}$$

During fluid splitting, no temperature change occurs as the inlet temperature remains identical across all outlets. Therefore, a single temperature node suffices to represent the entire splitting process. The flow rate relationship between the main loop and the branches of the entire intermediate loop in Fig. 1 can be given by:

$$\dot{m}_0 = \dot{m}_1 + \dot{m}_2. \tag{6}$$

(3) Heat Exchanger

While a detailed heat current model for transient processes in heat exchangers was deduced in prior work [21] to achieve high-precision

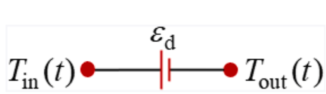


Fig. 2. Heat current model of a pipeline.

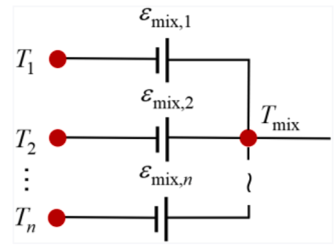


Fig. 3. Heat current model of fluid mixing processes.

simulation, its complexity hinders controller design. To this end, we propose a simplified model that captures the essential heat transfer and migration dynamics. Our model is based on the original steady-state heat current model for a counter-flow heat exchanger [20], as illustrated in Fig. 4(a). This foundational model comprises a thermal resistance R, along with *heat transfer thermal potentials* ε_{h} and ε_{c}:

$$\begin{cases} R = \dfrac{G_{\text{c}} e^{-KA/G_{\text{h}}} - G_{\text{h}} e^{-KA/G_{\text{c}}}}{G_{\text{h}} G_{\text{c}} \left(e^{-KA/G_{\text{h}}} - e^{-KA/G_{\text{c}}}\right)} \\ \varepsilon_x(t) = \dfrac{Q_x(t)}{G_x} (x = \text{c}, \text{h}) \end{cases} \tag{7}$$

where $G_{\text{c}} = \dot{m}_{\text{c}} c_{p,\text{c}}$ and $G_{\text{h}} = \dot{m}_{\text{h}} c_{p,\text{h}}$ denote the heat capacity rates of cold and hot fluids, respectively, and the corresponding fluid temperatures are: inlets T_{hin} and T_{cin}; outlets T_{hout} and T_{cout}. K and A are the heat transfer coefficient and area, respectively. This model lumps the nonlinearities of the heat transfer process into the thermal resistance expression, such that a linear thermal Ohm's law governs the relationship between the inlet temperature difference and the thermal resistance, thereby overcoming a key limitation of traditional methods.

To account for pipeline delays, we propose a modified model (Fig. 4(b)) by adding delay thermal potentials $\varepsilon_{\text{d,h}}$ and $\varepsilon_{\text{c,h}}$. Their expressions are derived from Eq. (2) and are applied to the hot and cold sides, respectively. Thus, the model effectively captures both the nonlinear effects of heat transfer and the delays induced by pipeline transport.

(4) Water Tank

The unified water tank temperature T_{T} varies in response to the inlet temperature T_{Ti} and the mass flow rate $\dot{m}_0$. Under the assumption that the specific heat capacity of water remains constant, the energy conservation principle yields the following constraint on the ratio of change of T_{T}

$$\dot{T}_{\text{T}}(t) = \frac{\dot{m}_0(t)[T_{\text{Ti}}(t) - T_{\text{T}}(t)]}{m_{\text{T}}(t)}. \tag{8}$$

Considering that the variation of T_{T} is affected by the temperature (difference) instead of heat flow, utilizing a temperature controlled thermal potential $\varepsilon_{\text{T}}(t)$ to describe T_{T} as shown in Fig. 5 might be more suitable than defining a thermal capacitance. To reflect the constraint of Eq. (8), the definition of $\varepsilon_{\text{T}}(t)$ can be given as

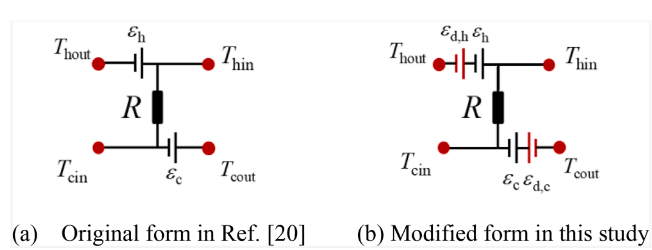


Fig. 4. Heat current model of a counter-flow heat exchanger.

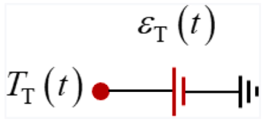


Fig. 5. Heat current model of the water tank.

$$\begin{cases} \varepsilon_{\mathrm{T}}(0) = T_{\mathrm{T}}(0) \\ \dot{\varepsilon}_{\mathrm{T}}(t) = \dfrac{\dot{m}_0(t)[T_{\mathrm{Ti}}(t) - T_{\mathrm{T}}(t)]}{m_{\mathrm{T}}(t)} \end{cases} \tag{9}$$

2.2. Heat current model of the system

Connecting components topologically yields the heat current model of the intermediate circulation loop as shown in Fig. 6. Here, MR1–3 represent models of AHE1, AHE2, and OHE, detailed in Fig. 4(b) for control. For distinction, hollow nodes represent open nodes in the system, which can admit external heat flow, whereas solid nodes represent the opposite. Additionally, the subscripts differentiating the *delay thermal potentials* of individual pipelines are omitted in the diagram for clarity. In practice, the delay time of each pipeline varies according to its respective length.

Since the heat current model takes the equivalent circuit form, applying the Kirchhoff's law the constraint equation for temperature can be derived directly from Kirchhoff's voltage law:

$$T_{\mathrm{hout1}} = T_{\mathrm{hin1}}(t - \tau_{\mathrm{H1}}) - \frac{1}{R_1 G_{\mathrm{h1}}}[T_{\mathrm{hin1}}(t - \tau_{\mathrm{H1}}) - T_{\mathrm{T}}(t - \tau_{\mathrm{a}})] \tag{10}$$

$$T_{\mathrm{hout2}} = T_{\mathrm{hin2}}(t - \tau_{\mathrm{H2}}) - \frac{1}{R_2 G_{\mathrm{h2}}}[T_{\mathrm{hin2}}(t - \tau_{\mathrm{H2}}) - T_{\mathrm{T}}(t - \tau_{\mathrm{b}})] \tag{11}$$

where $\tau_{\mathrm{a}}=\tau_{\mathrm{T},D}+\tau_{\mathrm{D,H1}}+\tau_{\mathrm{H1}}$ and $\tau_{\mathrm{b}}=\tau_{\mathrm{T},D}+\tau_{\mathrm{D,H2}}+\tau_{\mathrm{H2}}$ represent the time taken for the work mass to flow out of the water tank until it flows through the two gas-water heat exchangers, respectively.

The inlet temperature of the water tank T_{Ti}, can be obtained from Eqs. (1)~(3), (6) and (7). Considering the conservation of energy, the initial internal energy of the water tank and the combination of Eq. (8), it can be concluded that at this time the water temperature in the tank with the time change rule is as follows:

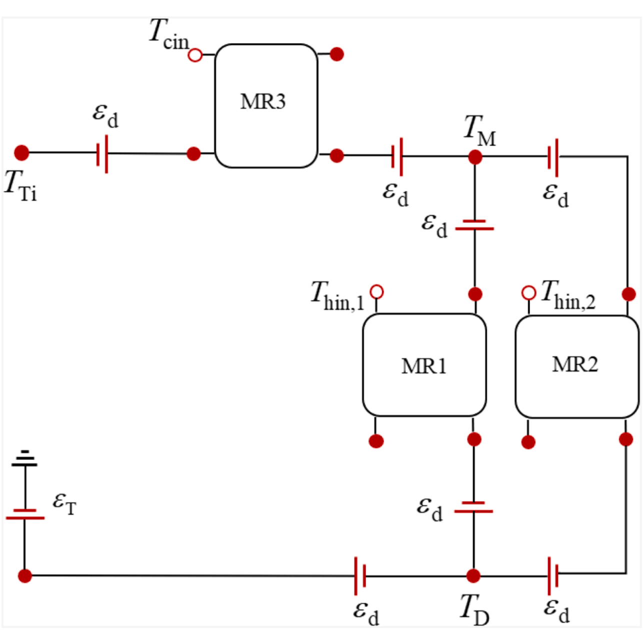


Fig. 6. Heat current model of the intermediate circulation loop.

$$\begin{aligned} \dot{T}_{\mathrm{T}} = \left[1 - \frac{1}{G_{\mathrm{h3}}R_3}\right] & \left\{ \begin{aligned} & \frac{\dot{m}_1}{m_{\mathrm{T}}}\left[T_{\mathrm{T}}(t-\tau_1) + \frac{T_{\mathrm{hin1}}(t-\tau_3) - T_{\mathrm{T}}(t-\tau_1)}{G_{\mathrm{h1}}R_1}\right] \\ & + \frac{\dot{m}_2}{m_{\mathrm{T}}}\left[T_{\mathrm{T}}(t-\tau_2) + \frac{T_{\mathrm{hin2}}(t-\tau_4) - T_{\mathrm{T}}(t-\tau_2)}{G_{\mathrm{h2}}R_2}\right] \end{aligned} \right\} \\ & + \frac{\dot{m}_0}{m_{\mathrm{T}}}\left[\frac{T_{\mathrm{cin3}}(t-\tau_5)}{G_{\mathrm{h3}}R_3} - T_{\mathrm{T}}\right] \end{aligned} \tag{12}$$

where $\tau_1 = \tau_{\mathrm{T},D} + \tau_{\mathrm{D,H1}} + \tau_{\mathrm{H1}} + \tau_{\mathrm{H1},M} + \tau_{\mathrm{M,H3}} + \tau_{\mathrm{H3}} + \tau_{\mathrm{H3,T}}$, $\tau_2 = \tau_{\mathrm{T},D} + \tau_{\mathrm{D,H2}} + \tau_{\mathrm{H2}} + \tau_{\mathrm{H2},M} + \tau_{\mathrm{M,H3}} + \tau_{\mathrm{H3}} + \tau_{\mathrm{H3,T}}$, $\tau_3 = \tau_{\mathrm{H1}} + \tau_{\mathrm{H1},M} + \tau_{\mathrm{M,H3}} + \tau_{\mathrm{H3}} + \tau_{\mathrm{H3,T}}$, $\tau_4 = \tau_{\mathrm{H2}} + \tau_{\mathrm{H2},M} + \tau_{\mathrm{M,H3}} + \tau_{\mathrm{H3}} + \tau_{\mathrm{H3,T}}$, and $\tau_5 = \tau_{\mathrm{H3}} + \tau_{\mathrm{H3,T}}$.

3. System control algorithm design

3.1. Predictive model

The primary control objective for the Intermediate Circulating Loop in Fig. 1 is to maintain the air outlet temperatures below their limits to protect the electronic components, while preventing the intermediate medium from boiling. For the MPC design, the two air outlet temperatures are selected as the controlled variables, forming the output vector $\boldsymbol{y}$ as:

$$\mathbf{y} = \begin{bmatrix} T_{hout1} - T^*_{hout1} \\ T_{hout2} - T^*_{hout2} \end{bmatrix}, \tag{13}$$

where the superscript '*' denotes the reference value, thus, $\boldsymbol{y}$ represents the offset from this reference. In practice, this reference could be set as a derated value below its specified limit.

Considering the adjustable factors in the system, the rates of change of the mass flow rates in the main circuit and in branch 1 are selected as the control variables, forming the input vector $\boldsymbol{u}$ as

$$\mathbf{u} = \begin{bmatrix} \ddot{m}_0 \\ \ddot{m}_1 \end{bmatrix}. \tag{14}$$

Herein, the time derivative of mass flow rates is designated as the direct control input, which is indirectly regulated through the coordinated adjustment of pumps and valves.

The state vector $\boldsymbol{x}$ is defined using the deviations of key state variables from their reference values. These variables include the water tank temperature, the main circuit flow rate, and the branch 1 flow rate:

$$\mathbf{x} = \begin{pmatrix} T_{\mathrm{T}} - T^*_{\mathrm{T}} \\ \dot{m}_0 - \dot{m}^*_0 \\ \dot{m}_1 - \dot{m}^*_1 \end{pmatrix}. \tag{15}$$

Based on the heat current model in Fig. 6 and the derived temperature expressions in Eqs. (10)-(12), a state-space representation is developed for the system. As the system exhibits time delays, the resulting equations incorporate time delay effects beyond those found in standard state-space models:

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{T}_{\mathrm{T}} - \dot{T}^*_{\mathrm{T}} \\ \ddot{m}_0 - \ddot{m}^*_0 \\ \ddot{m}_1 - \ddot{m}^*_1 \end{bmatrix} = \mathbf{A}_{c_{\mathrm{delay}}}\mathbf{x} + \mathbf{A}_{\tau_1}\mathbf{x}(t-\tau_1) + \mathbf{A}_{\tau_2}\mathbf{x}(t-\tau_2) + \mathbf{B}_{c_{\mathrm{delay}}}\mathbf{u}, \tag{16}$$

$$\mathbf{y} = \mathbf{C}_{c_{\mathrm{delay}}}\mathbf{x} + \mathbf{C}_{\tau_a}\mathbf{x}(t-\tau_a) + \mathbf{C}_{\tau_b}\mathbf{x}(t-\tau_b), \tag{17}$$

where

$$\mathbf{A}_{c_{\mathrm{delay}}} = \begin{bmatrix} \dfrac{\partial \dot{T}_{\mathrm{T}}}{\partial T_{\mathrm{T}}} & \dfrac{\partial \dot{T}_{\mathrm{T}}}{\partial \dot{m}_0} & \dfrac{\partial \dot{T}_{\mathrm{T}}}{\partial \dot{m}_1} \\ \dfrac{\partial \ddot{m}_0}{\partial T_{\mathrm{T}}} & \dfrac{\partial \ddot{m}_0}{\partial \dot{m}_0} & \dfrac{\partial \ddot{m}_0}{\partial \dot{m}_1} \\ \dfrac{\partial \ddot{m}_1}{\partial T_{\mathrm{T}}} & \dfrac{\partial \ddot{m}_1}{\partial \dot{m}_0} & \dfrac{\partial \ddot{m}_1}{\partial \dot{m}_1} \end{bmatrix}, \tag{18}$$

$$\mathbf{A}_{\tau_1} = \begin{bmatrix} \dfrac{\partial \dot{T}_{\mathrm{T}}}{\partial T_{\mathrm{T}}(t-\tau_1)} & \dfrac{\partial \dot{T}_{\mathrm{T}}}{\partial \dot{m}_0(t-\tau_1)} & \dfrac{\partial \dot{T}_{\mathrm{T}}}{\partial \dot{m}_1(t-\tau_1)} \\ \dfrac{\partial \ddot{m}_0}{\partial T_{\mathrm{T}}(t-\tau_1)} & \dfrac{\partial \ddot{m}_0}{\partial \dot{m}_0(t-\tau_1)} & \dfrac{\partial \ddot{m}_0}{\partial \dot{m}_1(t-\tau_1)} \\ \dfrac{\partial \ddot{m}_1}{\partial T_{\mathrm{T}}(t-\tau_1)} & \dfrac{\partial \ddot{m}_1}{\partial \dot{m}_0(t-\tau_1)} & \dfrac{\partial \ddot{m}_1}{\partial \dot{m}_1(t-\tau_1)} \end{bmatrix}, \tag{19}$$

$$\mathbf{A}_{\tau_2} = \begin{bmatrix} \dfrac{\partial \dot{T}_{\mathrm{T}}}{\partial T_{\mathrm{T}}(t-\tau_2)} & \dfrac{\partial \dot{T}_{\mathrm{T}}}{\partial \dot{m}_0(t-\tau_2)} & \dfrac{\partial \dot{T}_{\mathrm{T}}}{\partial \dot{m}_1(t-\tau_2)} \\ \dfrac{\partial \ddot{m}_0}{\partial T_{\mathrm{T}}(t-\tau_2)} & \dfrac{\partial \ddot{m}_0}{\partial \dot{m}_0(t-\tau_2)} & \dfrac{\partial \ddot{m}_0}{\partial \dot{m}_1(t-\tau_2)} \\ \dfrac{\partial \ddot{m}_1}{\partial T_{\mathrm{T}}(t-\tau_2)} & \dfrac{\partial \ddot{m}_1}{\partial \dot{m}_0(t-\tau_2)} & \dfrac{\partial \ddot{m}_1}{\partial \dot{m}_1(t-\tau_2)} \end{bmatrix}, \tag{20}$$

$$\mathbf{B}_{c_{\mathrm{delay}}} = \begin{bmatrix} \dfrac{\partial \dot{T}_{\mathrm{T}}}{\partial \dot{m}_0} & \dfrac{\partial \dot{T}_{\mathrm{T}}}{\partial \dot{m}_1} \\ \dfrac{\partial \ddot{m}_0}{\partial \dot{m}_0} & \dfrac{\partial \ddot{m}_0}{\partial \dot{m}_1} \\ \dfrac{\partial \ddot{m}_1}{\partial \dot{m}_0} & \dfrac{\partial \ddot{m}_1}{\partial \dot{m}_1} \end{bmatrix}, \tag{21}$$

$$\mathbf{C}_{c_{\mathrm{delay}}} = \begin{bmatrix} \dfrac{\partial T_{\mathrm{hout1}}}{\partial T_{\mathrm{T}}} & \dfrac{\partial T_{\mathrm{hout1}}}{\partial \dot{m}_0} & \dfrac{\partial T_{\mathrm{hout1}}}{\partial \dot{m}_1} \\ \dfrac{\partial T_{\mathrm{hout2}}}{\partial T_{\mathrm{T}}} & \dfrac{\partial T_{\mathrm{hout2}}}{\partial \dot{m}_0} & \dfrac{\partial T_{\mathrm{hout2}}}{\partial \dot{m}_1} \end{bmatrix}, \tag{22}$$

$$\mathbf{C}_{\tau_a} = \begin{bmatrix} \dfrac{\partial T_{\mathrm{hout1}}}{\partial T_{\mathrm{T}}(t-\tau_a)} & \dfrac{\partial T_{\mathrm{hout1}}}{\partial \dot{m}_0(t-\tau_a)} & \dfrac{\partial T_{\mathrm{hout1}}}{\partial \dot{m}_1(t-\tau_a)} \\ \dfrac{\partial T_{\mathrm{hout2}}}{\partial T_{\mathrm{T}}(t-\tau_a)} & \dfrac{\partial T_{\mathrm{hout2}}}{\partial \dot{m}_0(t-\tau_a)} & \dfrac{\partial T_{\mathrm{hout2}}}{\partial \dot{m}_1(t-\tau_a)} \end{bmatrix}, \tag{23}$$

$$\mathbf{C}_{\tau_b} = \begin{bmatrix} \dfrac{\partial T_{\mathrm{hout1}}}{\partial T_{\mathrm{T}}(t-\tau_{\mathrm{b}})} & \dfrac{\partial T_{\mathrm{hout1}}}{\partial \dot{m}_0(t-\tau_{\mathrm{b}})} & \dfrac{\partial T_{\mathrm{hout1}}}{\partial \dot{m}_1(t-\tau_{\mathrm{b}})} \\ \dfrac{\partial T_{\mathrm{hout2}}}{\partial T_{\mathrm{T}}(t-\tau_{\mathrm{b}})} & \dfrac{\partial T_{\mathrm{hout2}}}{\partial \dot{m}_0(t-\tau_{\mathrm{b}})} & \dfrac{\partial T_{\mathrm{hout2}}}{\partial \dot{m}_1(t-\tau_{\mathrm{b}})} \end{bmatrix}, \tag{24}$$

The system equations contain multiple delay terms due to varying delay times throughout the system. Given that the delay difference between the two branches is insignificant ($\tau_1 \approx \tau_2$), the equations can be simplified for ease of derivation. This simplification is justified because the MPC incorporates feedback regulation, which can compensate for the inaccuracies introduced by this assumption. Therefore, the state variables at different times are reduced, and Eqs. (16) and (17) can transferred into:

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{T}_{\mathrm{T}} - \dot{T}_{\mathrm{T}}^{*} \\ \ddot{m}_0 - \ddot{m}_0^{*} \\ \ddot{m}_1 - \ddot{m}_1^{*} \end{bmatrix} = \mathbf{A}_{c_{\mathrm{delay}}}\mathbf{x} + \mathbf{A}_{\tau_2}\mathbf{x}(t-\tau_2) + \mathbf{B}_{c_{\mathrm{delay}}}\mathbf{u}, \tag{25}$$

$$\mathbf{y} = \begin{bmatrix} T_{hout1} - T_{hout1}^{*} \\ T_{hout2} - T_{hout2}^{*} \end{bmatrix} = \mathbf{C}_{c_{\mathrm{delay}}}\mathbf{x} + \mathbf{C}_{\tau_b}\mathbf{x}(t-\tau_b). \tag{26}$$

At this point, the expressions for the 36 partial derivatives of the elements in the Eqs. (18), (20), (21), (22) and (24) are given in the Appendix A.

In order for MPC to be used for practical applications, the above continuous model is converted to a discrete model, which is subjected to a forward Eulerian discretization

$$\dot{\mathbf{x}} = \frac{1}{\Delta t}(\mathbf{x}_{k+1} - \mathbf{x}_k), \tag{27}$$

where Δt is the time step and the subscript k is an arbitrary step.

The discrete state space expression is obtained:

$$\mathbf{x}_{k+1} = \left(\mathbf{I} + \Delta t\mathbf{A}_{c_{\mathrm{delay}}}\right)\mathbf{x}_k + \Delta t\mathbf{A}_{\tau 2}\mathbf{x}_{k-b_1} + \Delta t\mathbf{B}_{c_{\mathrm{delay}}}u_k, \tag{28}$$

$$\mathbf{y}_k = \mathbf{C}_{c_{\mathrm{delay}}}\mathbf{x}_k + \mathbf{C}_{\tau_b}\mathbf{x}_{k-b_2}, \tag{29}$$

where $b_1 = \tau_2/\Delta t$ and $b_2 = \tau_{\mathrm{b}}/\Delta t$. In practice, b_1 and b_2 can be taken to be an integer when the time step is taken sufficiently small. We know that b_1 is always greater than b_2 by physical relationship, so in the case of $b_1 > b_2$ for the time domain expansion, to obtain the following system of generalized equations:

$$\mathbf{x}_{d_{k+1}} = \mathbf{A}\mathbf{x}_{d_k} + \mathbf{B}u_k, \tag{30}$$

$$\mathbf{y}_k = \mathbf{C}\mathbf{x}_{d_k}, \tag{31}$$

where

$$\mathbf{x}_{d_{k+1}} = \begin{bmatrix} \mathbf{x}_{k+1} \\ \mathbf{x}_k \\ \vdots \\ \mathbf{x}_{k-b_1+1} \end{bmatrix}, \tag{32}$$

$$\mathbf{A} = \begin{bmatrix} \mathbf{I} + \Delta t\mathbf{A}_{c_{\mathrm{delay}}} & \mathbf{0} & \cdots & \mathbf{0} & \Delta t\mathbf{A}_{\tau_2} \\ \mathbf{I} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} & \cdots & \mathbf{0} \\ \vdots & & \ddots & & \vdots \\ \mathbf{0} & \cdots & \mathbf{0} & \mathbf{I} & \mathbf{0} \end{bmatrix}, \tag{33}$$

$$\mathbf{x}_{d_k} = \begin{bmatrix} \mathbf{x}_k \\ \mathbf{x}_{k-1} \\ \vdots \\ \mathbf{x}_{k-b_1} \end{bmatrix}, \tag{34}$$

$$\mathbf{B} = \begin{bmatrix} \Delta t\mathbf{B}_{c_{\mathrm{delay}}} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}, \tag{35}$$

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_{c_{\mathrm{delay}}} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{C}_{\tau_b} & \mathbf{0} & \cdots & \mathbf{0} \end{bmatrix}. \tag{36}$$

3.2. Control design

To maintain stability and performance despite modeling errors and disturbances, a feedback correction based on the control error is introduced before the MPC design to eliminate steady-state errors. The corresponding control structure is depicted in Fig. 7:

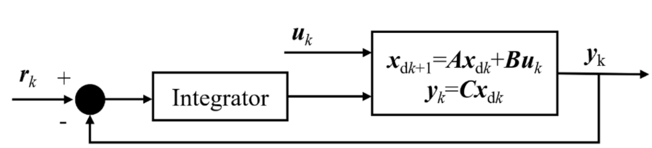


Fig. 7. Schematic of Feedback Control.

Using a forward Euler integrator, the output of the integrator is:

$$\mathbf{e}_{i_{k+1}} = \mathbf{e}_{i_k} + \Delta t[\boldsymbol{r}_k - \boldsymbol{y}_k] = \mathbf{e}_{i_k} + \Delta t\left[\boldsymbol{r}_k - \mathbf{C}\mathbf{x}_{d_k}\right] = -\Delta t\mathbf{C}\mathbf{x}_{d_k} + \mathbf{e}_{i_k} + \Delta t\boldsymbol{r}_k \tag{37}$$

where $\boldsymbol{r}_k$ is the reference signal vector, and $\boldsymbol{e}$ is the control deviation vector.

In this case, the system is linearized around a trim point, and the reference signal $\boldsymbol{r}_k$ in the figure is zero. Combining the deviations with those of the original model yields a state space expression for the whole system:

$$\widetilde{\mathbf{x}}_{k+1} = \begin{bmatrix} \mathbf{x}_{d_{k+1}} \\ \mathbf{e}_{i_{k+1}} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ -\Delta t\mathbf{C} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{d_k} \\ \mathbf{e}_{i_k} \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ 0 \end{bmatrix} \mathbf{u}_k = \widetilde{\mathbf{A}}\widetilde{\mathbf{x}}_k + \widetilde{\mathbf{B}}u_k \tag{38}$$

$$\widetilde{\mathbf{y}}_k = \begin{bmatrix} y_k \\ \mathbf{e}_{i_k} \end{bmatrix} = \begin{bmatrix} \mathbf{C} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{d_k} \\ \mathbf{e}_{i_k} \end{bmatrix} = \widetilde{\mathbf{C}}\widetilde{\mathbf{x}}_k \tag{39}$$

In the MPC controller part, the objective is to find the optimal control set:

$$\mathbf{U} = [\,\mathbf{u}_0 \quad \dots \quad \mathbf{u}_{N-1}\,] \tag{40}$$

The controller aims to balance tracking accuracy, control smoothness, and system stability. This balance, which optimizes both dynamic and steady-state performance, is achieved by tuning the weight matrices in the cost function:

$$\mathrm{J}\left(\widetilde{\mathbf{x}}_0^T, \mathbf{U}\right) = \sum_{k=0}^{N-1} \left[\widetilde{\mathbf{x}}_k^T \mathbf{Q}\widetilde{\mathbf{x}}_k + \mathbf{u}_k^T \mathbf{R}\mathbf{u}_k\right] + \widetilde{\mathbf{x}}_N^T \mathbf{P}\widetilde{\mathbf{x}}_N \tag{41}$$

where N is the number of time steps for the prediction, $\boldsymbol{Q}$ and $\boldsymbol{P}$ are the state weight matrices, which are usually semi-positive definite matrices, and $\boldsymbol{R}$ is the input weight matrix, which is usually positive definite matrices.

Expanding Eq. (41) yields:

$$J(\widetilde{\mathbf{x}}_0, \mathbf{U}) = \widetilde{\mathbf{x}}_0^T \mathbf{Q}\widetilde{\mathbf{x}}_0 + [\,\widetilde{\mathbf{x}}_1^T \quad \widetilde{\mathbf{x}}_2^T \quad \cdots \quad \widetilde{\mathbf{x}}_N^T\,]\overline{\mathbf{Q}} \begin{bmatrix} \widetilde{\mathbf{x}}_1^T \\ \widetilde{\mathbf{x}}_2^T \\ \vdots \\ \widetilde{\mathbf{x}}_N^T \end{bmatrix} + [\,\mathbf{u}_0 \quad \mathbf{u}_1 \quad \cdots \quad \mathbf{u}_{N-1}\,]\overline{\mathbf{R}} \begin{bmatrix} \mathbf{u}_0 \\ \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_{N-1} \end{bmatrix} \tag{42}$$

where

$$\overline{\mathbf{Q}} = \begin{bmatrix} \mathbf{Q} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{Q} & \mathbf{0} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \mathbf{Q} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{P} \end{bmatrix} \tag{43}$$

$$\overline{\mathbf{R}} = \begin{bmatrix} \mathbf{R} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{R} & \mathbf{0} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \mathbf{R} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{R} \end{bmatrix} \tag{44}$$

Using Eq. (38), future states can be predicted for the next few steps:

$$\begin{cases} \widetilde{\mathbf{x}}_1 = \widetilde{\mathbf{A}}\widetilde{\mathbf{x}}_0 + \mathbf{B}\mathbf{u}_0 \\ \widetilde{\mathbf{x}}_2 = \widetilde{\mathbf{A}}\widetilde{\mathbf{x}}_1 + \widetilde{\mathbf{B}}\mathbf{u}_1 = \widetilde{\mathbf{A}}^2\widetilde{\mathbf{x}}_0 + \widetilde{\mathbf{A}}\widetilde{\mathbf{B}}\mathbf{u}_0 + \widetilde{\mathbf{B}}\mathbf{u}_1 \\ \dots \\ \widetilde{\mathbf{x}}_N = \widetilde{\mathbf{A}}^N\widetilde{\mathbf{x}}_0 + \widetilde{\mathbf{A}}^{N-1}\widetilde{\mathbf{B}}\mathbf{u}_0 + \widetilde{\mathbf{A}}^{N-2}\widetilde{\mathbf{B}}\mathbf{u}_1 + \cdots + \widetilde{\mathbf{B}}\mathbf{u}_{N-1} \end{cases} \tag{45}$$

Based on the prediction model, the sequence of states within N steps is written as:

$$\begin{bmatrix} \widetilde{\mathbf{x}}_1 \\ \widetilde{\mathbf{x}}_2 \\ \vdots \\ \widetilde{\mathbf{x}}_N \end{bmatrix} = \begin{bmatrix} \widetilde{\mathbf{B}} & \mathbf{0} & \cdots & \mathbf{0} \\ \widetilde{\mathbf{A}}\widetilde{\mathbf{B}} & \widetilde{\mathbf{B}} & \cdots & \mathbf{0} \\ \vdots & \vdots & \cdots & \vdots \\ \widetilde{\mathbf{A}}^{N-1}\widetilde{\mathbf{B}} & \widetilde{\mathbf{A}}^{N-2}\widetilde{\mathbf{B}} & \cdots & \widetilde{\mathbf{B}} \end{bmatrix} \begin{bmatrix} \mathbf{u}_0 \\ \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_{N-1} \end{bmatrix} + \begin{bmatrix} \widetilde{\mathbf{A}} \\ \widetilde{\mathbf{A}}^2 \\ \vdots \\ \widetilde{\mathbf{A}}^N \end{bmatrix} \widetilde{\mathbf{x}}_0 = \mathbf{S}\mathbf{U} + \mathbf{W}\widetilde{\mathbf{x}}_0 \tag{46}$$

Substituting Eq. (46) into Eq. (42) and simplifying yields:

$$J\left(\widetilde{\mathbf{x}}_0^T, \mathbf{U}\right) = \frac{1}{2}\widetilde{\mathbf{x}}_0^T \mathbf{Y}\widetilde{\mathbf{x}}_0 + \frac{1}{2}\mathbf{U}^T \mathbf{H}\mathbf{U} + \widetilde{\mathbf{x}}_0^T \mathbf{F}\mathbf{U} \tag{47}$$

where $\mathbf{Y} = 2\mathbf{Q} + 2\mathbf{W}^T\overline{\mathbf{Q}}\mathbf{W}, \mathbf{H} = 2\mathbf{S}^\mathbf{T}\overline{\mathbf{Q}}\mathbf{S} + 2\overline{\mathbf{R}}, \mathbf{F} = 2\mathbf{W}^\mathbf{T}\overline{\mathbf{Q}}\mathbf{S}$.

The optimal control should have

$$\frac{\partial J\left(\widetilde{\mathbf{x}}_0^T, \mathbf{U}\right)}{\partial \mathbf{U}} = \mathbf{H}\mathbf{U} + \mathbf{F}^T\widetilde{\mathbf{x}}_0 = 0 \tag{48}$$

The optimal control solution can be found analytically as

$$\mathbf{U} = -\mathbf{H}^{-1}\left(\mathbf{F}^T\widetilde{\mathbf{x}}_0\right) \tag{49}$$

In order to adapt to the dynamic changes and uncertainties of the system and to ensure the flexibility and robustness of the control strategy, the optimal control is executed in a single step, i.e., a rolling approach can be used to obtain the optimal control scheme under changing boundary conditions.

Since the variation in the flow rate change rate results from adjustments to the pump valve—a process that requires a certain amount of time—there exists an upper limit on how quickly the flow rate can change. Constrained MPC takes into account restrictions on the system's inputs and/or outputs. When such constraints apply to the system output, they can be expressed in the following form:

$$\mathbf{u}_{k\min} \le \mathbf{u}_k \le \mathbf{u}_{k\max}, k = 0, 1, \dots, N-1 \tag{50}$$

Combining with Eq. (40), it becomes:

$$\mathbf{A}_{\text{constraint}}\mathbf{U} \le \mathbf{b}_{\text{constraint}} \tag{51}$$

where

$$\mathbf{A}_{\text{constraint}} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ -1 & 0 & 0 & \cdots & 0 \\ -1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 0 & \cdots & 0 \\ 0 & -1 & 0 & \cdots & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ \vdots & \vdots & 0 & \ddots & 0 \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 0 & -1 \\ 0 & 0 & \cdots & 0 & -1 \end{bmatrix} \tag{52}$$

$$\mathbf{b}_{\text{constraint}} = \begin{bmatrix} \mathbf{u}_{0\max} \\ -\mathbf{u}_{0\min} \\ \mathbf{u}_{1\max} \\ -\mathbf{u}_{1\min} \\ \vdots \\ \mathbf{u}_{N-1_{\max}} \\ -\mathbf{u}_{N-1_{\min}} \end{bmatrix} \tag{53}$$

After obtaining the constraints, they are placed in the controller, which is mainly expressed in the "quadprog function" in MATLAB, and optimized to solve the Quadratic Programming (QP) problem.

3.3. Calculation process

By transforming the heat current model into a state-space expression model and combining it with feedback control, a constrained MPC approach is obtained, and then the control algorithm of the whole FTMS of aircraft engines is obtained by taking into account the control requirements of the actual system and the system characteristics. Finally, combined with the temperature and flow information collected by the sensor, the automatic control of the whole system can be accomplished, and its calculation flow is shown in Fig. 8. In normal operating conditions, the direct use of the model predictive controller to calculate the pump frequency, valve opening instructions, but in practice, because there will be another mode, this time to switch to another controller, through the program setting control. Firstly, determine whether the water temperature at each point in the system exceeds the temperature limit, if so, carry out the forced heat exchange cooling process; secondly, determine whether there is a flow of air No 1 (in view of the sensor's possible measurement error, here to obtain the lower limit of 0.01kg/s), if there is no flow of water, then the valve will be shut down 1; finally, process the collected temperature and flow rate information, and then get the system prediction model at this time to optimize and solve and single-step implementation, according to the flow rate and pump valve correspondence output pump frequency, valve opening degree of the command, so as to realize the whole test bench automatic control, the actual controller behind is also based on the process design of this section.

4. Characterization of test benches

A full-scale ground-test experimental system, configured as shown in Fig. 1, has been constructed to simulate real flight conditions. Fig. 9 presents the sensor layout within the Intermediate Circulation Loop, where the circulating pump (CP) and regulating valves (V1, V2) are adjustable to achieve flow control. Temperature is measured at points 1∼14, while mass flow rates are monitored at locations A∼E. A

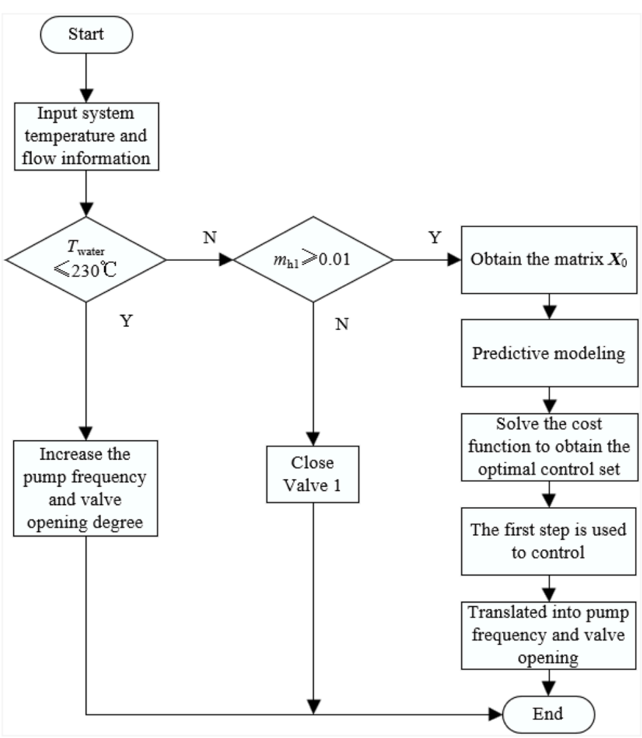


Fig. 8. Automatic control algorithm calculation flowchart.

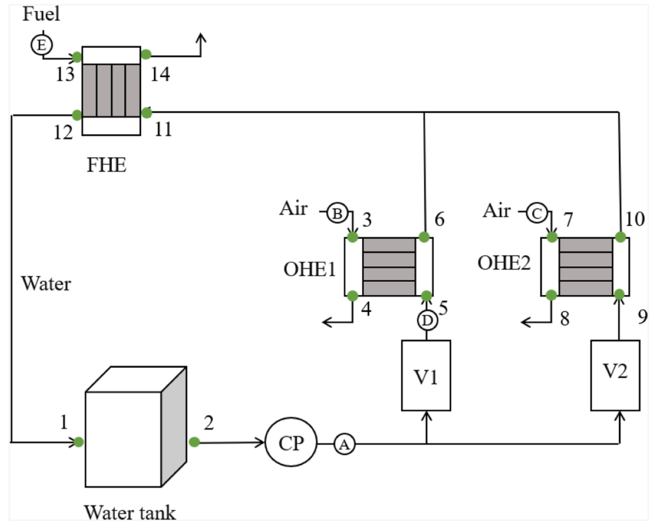


Fig. 9. Sensor layout within the Intermediate Circulation Loop.

photograph of the experimental setup and its key components is provided in Fig. 10.

The key characteristics to be determined for the experimental system encompass the following: water tank dynamic capacity, heat exchanger heat transfer properties, delays, and pump-valve flow regulation.

4.1. Water tank dynamic capacity identification

In order to get the dynamic capacity of the tank, a section of data with stable change in the outlet temperature of the water tank was selected, and according to Eq. (8), the temperature change rate of the water tank was obtained by fitting using the least squares method under the condition of known water tank inlet and outlet water temperatures and flow rates to get the current dynamic capacity of the water tank.

As shown in Fig. 11 is the inlet and outlet temperatures and flow rates of the water tank. Under a particular experiment, as the water flow rate in the loop becomes larger twice (around 50 s and 250 s) and the fuel inlet temperature in the fuel heat exchanger increases (at around 250 s) resulting in two large changes in the water tank, it can be divided into three segments for analysis: (1) 0 – 45 s, (2) 100 – 240 s, (3) 250 – 300 s.

During this experiment, the air flow rates of AHE1 and AHE2 were respectively stabilized at 0.80 kg/s and 0.24 kg/s; the fuel flow rate was stable at 1.30 kg/s from 0 to 240 s, and then slowly increased to 1.38 kg/s. In the intermediate circulation loop, the main water flow rate changed from 0.37 kg/s to 0.54 kg/s and finally stabilized at 0.65 kg/s at 40 s and 240 s respectively, while the branch 1 flow rate changed from 0.19 kg/s to 0.27 kg/s and finally stabilized at 0.32 kg/s. The inlet temperature of No 1 air slowly rose from 578 K to 700 K; the inlet temperature of No 2 air slowly rose from 515 K to 600 K; the fuel inlet temperature slowly rose from 303 K to 311 K and then quickly reached 381 K. After linear fitting (see Fig. 12), the temperature change rate of the storage tank was obtained, thereby calculating the dynamic flow rate of the storage tank. The dynamic flow rate calculation results were 11.69 kg, 11.08 kg, and 11.21 kg respectively. Combining the time lengths of each period, an average capacity can be calculated proportionally. The average of the above results is estimated as 11.22 kg.

4.2. Identification of heat transfer characteristics of heat exchangers

The calculation formula of thermal resistance in Eq. (7) relies on the identification of thermal conductivity, which can be calculated from the logarithmic mean temperature difference of each heat exchanger thermal conductivity. In the formula, the heat transfer heat flow Q is

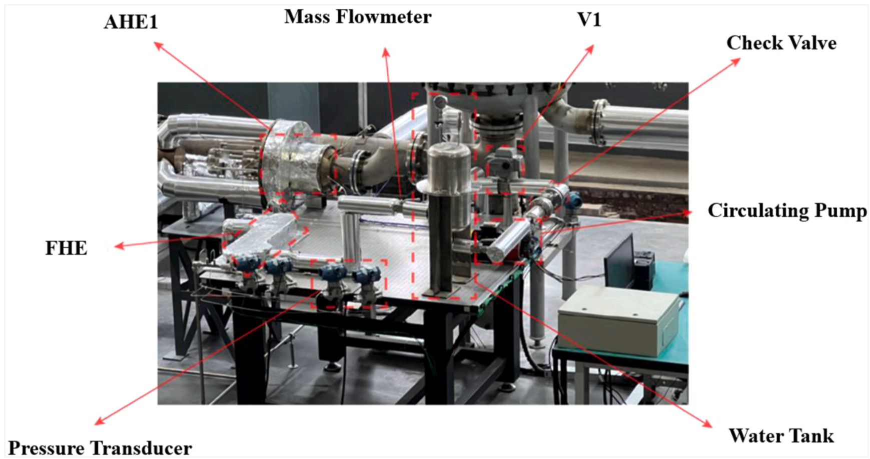


Fig. 10. A photograph of the experimental setup and its main components.

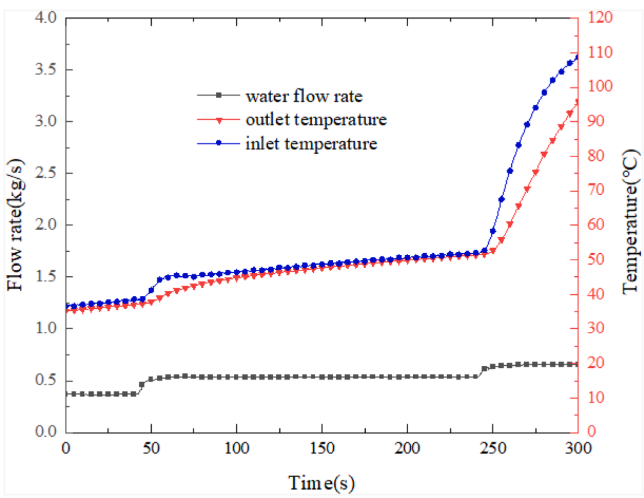


Fig. 11. Inlet and outlet temperatures and flow rates of the water tank.

obtained by taking the calculation of the average internal energy change of the fluid on both sides for a given constant pressure specific heat of the fluid ($c_{p,\mathrm{water}}$=3100 J/kg/K, $c_{p,\mathrm{air}}$=1050 J/kg/K and $c_{p,\mathrm{fuel}}$=2300 J/kg/K).

The operation data during the relatively smooth operation in the experiment were selected to obtain the inlet and outlet and flow rate data of each heat exchanger at this time, see Table 2. The table compares the thermal conductivities of each heat exchanger calculated from the data of this experiment with those calculated from the steady-state experiment data. The thermal conductivities are 713.7, 1276.1 and 11,409.2 W/K.

4.3. Delay identification

In order to analyze the delay characteristics τ of fluid flow and heat transfer in the system, this paper selects a dynamic experiment process that the cases of stepwise upward and downward adjustment of fuel inlet temperature, respectively, and extracts the peak and valley response times of the temperatures at each measuring point in the intermediate circulation loop, as shown in Table 3. In this experiment, the flow rate of the main water-side loop is 0.5 kg/s and branch 1 flow rate at 0.324 kg/s. The flow rates on the non-water side of each heat exchangers $\dot{m}_{\mathrm{h1}}$, $\dot{m}_{\mathrm{h2}}$ and $\dot{m}_{\mathrm{c}}$ are 0.503 kg/s, 0.25 kg/s, and 1.327 kg/s, respectively. Under the condition that the water flow rate remains constant, by analyzing the peak and valley values of the temperature responses at each measurement point, the delay of heat transfer within the system can be identified. Multiple extraction results show that the delay time at different positions in the system ranges from 30 s to 90 s, depending on factors such as the flow path of the working medium, local flow rate, and the coupling structure of the heat exchanger. Considering the typical response characteristics and the requirements of control design, this paper temporarily takes 60 s as the equivalent delay time of the system, which is used for the time-domain expansion of the state-space expression in the model predictive controller.

4.4. Characteristic fitting of pump and valve regulation for flow rate

There are three adjustable pump/valves on the lab bench: the variable frequency pump on the main line, pneumatic V1 on branch 1, and pneumatic V2 on branch 2. By adjusting these values, we can change the flow rate $\dot{m}_0$ in the main line, $\dot{m}_1$ in branch 1, and $\dot{m}_2$ in branch 2. Since the number of adjustable variables exceeds the number of controlled variables, in the experiment, under the condition that the opening degree c_2 of pneumatic V2 is given, w and c_1 were selected for joint regulation. The flow rate calculated by the controller needs to be converted into pump frequency and valve opening degree to achieve the control of flow rate. Therefore, it is necessary to find the regulation characteristics of pump and valve on flow rate. Here, the neural network method is used to directly fit the relationship between frequency w, opening degree c_1 and flow rate $\dot{m}_0$, $\dot{m}_1$. The pump valve flow information at $c_2 = 1$ was used for the fitting, and orthogonal sampling was used, with samples taken at 0.1 intervals. A total of 34 sets of data were selected with openings in the range of 0.4 to 1 and frequencies in the range of 18 to 140 Hz, and the fit is shown in Fig. 13. Due to the limited data collected at small openings, the fitting effect is poorer than that of w. The fit is shown in Table 4. This neural network has two hidden layers and ten neurons.

5. Experimental results and analysis

Based on these identified parameters in Section 4, a complete MPC controller was constructed and integrated into the experimental system using the software LabVIEW. In this section, we will conduct control experiments based on actual flight conditions.

5.1. Experimental conditions

Since frequent changes in operating conditions cause drastic fluctuations in heat loads, making the FTMS of aircraft engines face complex control challenges such as nonlinearity, strong coupling, dynamic constraints, etc. Here we simulate the switching of operating conditions that

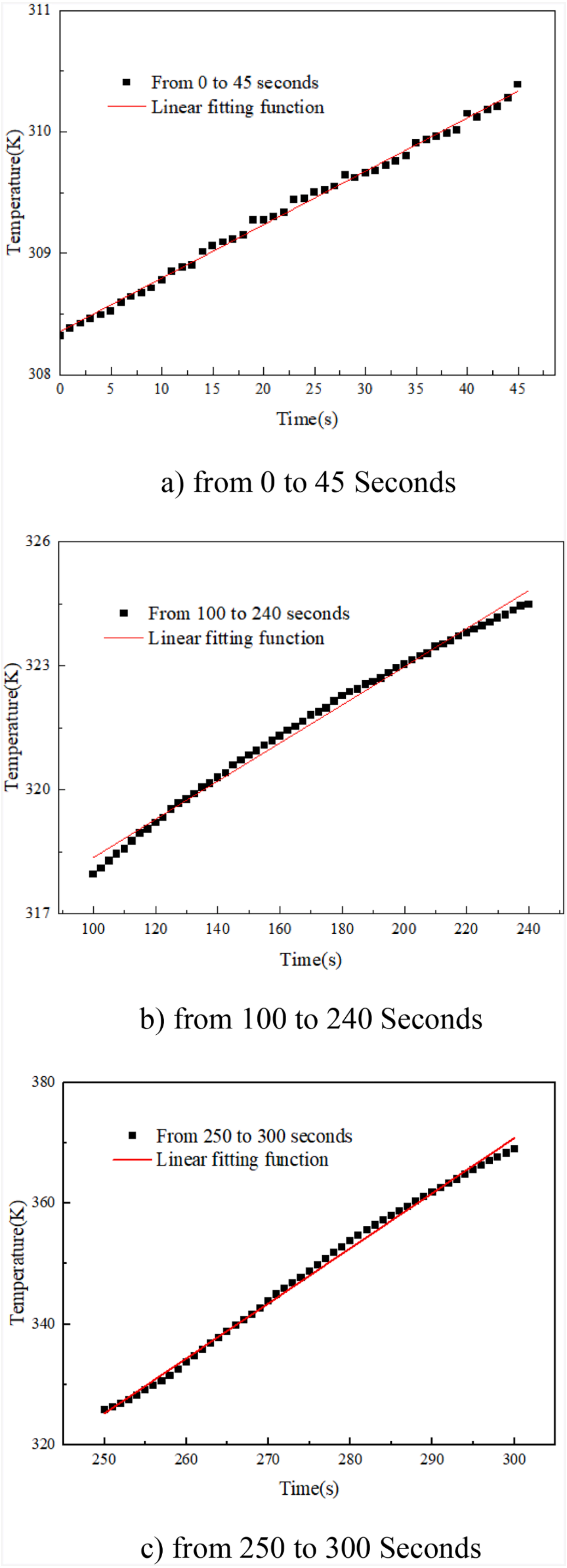


Fig. 12. Curve of Outlet Temperature of the Water Tank.

Table 2
The boundary temperatures, flow rates and the thermal conductivity of heat exchangers.

	AHE1	AHE2	FHE
Cold side inlet /K	373.7	373.7	352.4
Cold side outlet /K	473.8	444.4	432.8
Hot side inlet /K	741.5	662.0	459.8
Hot side outlet /K	568.3	404.6	375.3
Cold side flow rate /(kg/s)	0.372	0.393	1.450
Hot side flow rate /(kg/s)	0.650	0.442	0.765
Logarithmic mean temperature difference/K	229.211	95.65	24.89
Thermal conductivity/(W/K)	713.7	1276.1	11,409.2

Table 3
The time difference between peak and valley values of each pipeline section in the system.

pipeline section	peak time difference/s	valley time difference/s
$\tau_{T,D}+\tau_{D,H1}$	−36.87	5.51
$\tau_{T,D}+\tau_{D,H2}$	−0.1	7.74
τ_{H1}	1.78	−20.55
τ_{H2}	−1.04	10.93
$\tau_{H1,M}+\tau_{M,H3}+\tau_{H3}+\tau_{H3,T}$	−44.02	5.63
$\tau_{H2,M}+\tau_{M,H3}+\tau_{H3}+\tau_{H3,T}$	−77.76	−28.08

the aircraft will face in flight. Under strong time-varying conditions, the adjustment of the aircraft engine FTMS is simulated in two different flight states: a high Mach number low speed state and a ramjet cruise state. The boundary conditions are varied as shown in Table 5.

In the table, condition 1 and 3 indicate the high Mach number low-speed state, and condition 2 indicates the rammed cruise state.

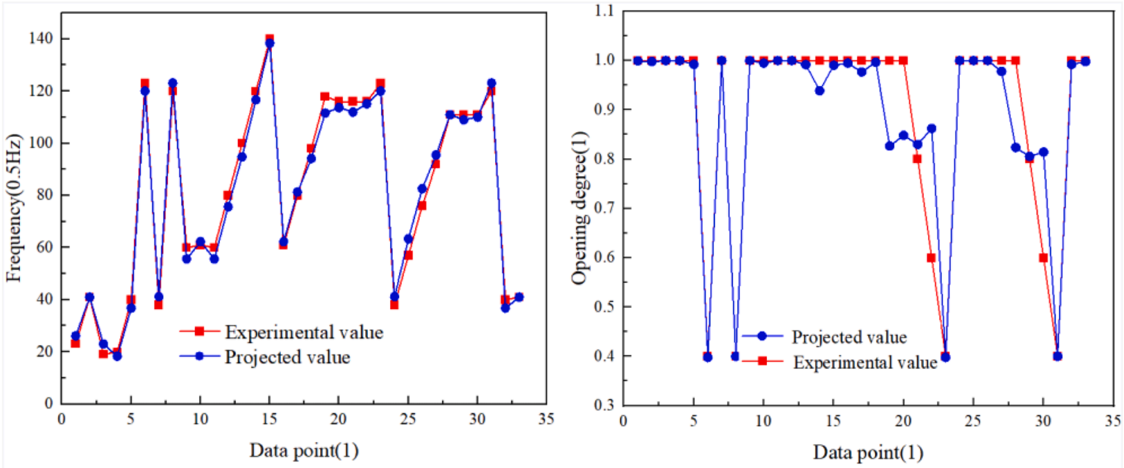
In order to raise the operating temperature of the system, it is necessary to increase the pressure by filling inert gases (such as argon) to raise the boiling point of water and prevent it from vaporizing. The boiling point of water after pressurization to 3 MPa was obtained during the experiments to be around 230 °C because of the requirement that the intermediate circulating work mass does not boil. That is, as long as we ensure that the temperature of the work material water in the intermediate circulating system during the experiment is below 503 K, water vapor will not be generated, which turns into a two-phase fluid, affecting the heat transfer characteristics, increasing the difficulty of calculation, and so on. The flow rate calculated by the control algorithm is converted through the relationship between pump frequency, valve opening and flow rate obtained by the flow identification device, and then the valve opening is controlled accordingly. The rest of the identified parts are in accordance with the results identified in Section III: the dynamic capacity of the water tank is set to 11.22 kg; the heat exchanger thermal conductivities are set to 713.7 W/K, 1276.1 W/K and 11,409.2 W/K, respectively; and τ_2 is set to 60 s.

In these states, the main fuel flow rate is about 1.29 kg/s, and the flow into the afterburner is about 0.29 kg/s with a temperature of 397 K. In this paper, the air outlet temperature needs to be stabilized at 400 K without boiling the material in the intermediate circulation system.

5.2. Experimental results

As shown in Fig. 14 for the strong time-varying experiments at various points of the test bench water temperature, the highest water temperature in the figure reaches about 500 K, the critical boiling (green line, this is AHE2's water-side outlet temperature), to achieve the control requirements of water does not boil. Table 6 shows the corresponding working conditions at different time periods.

As shown in Fig. 15–17, the inlet and outlet temperatures of each heat exchanger and the flow rates on both sides during the strong time-varying test are presented. In the figures, the fluid flow rates on both sides of the heat exchangers are indicated below the legends, with the flow rates



(a) The fit of w (b) The fit of c_1

Fig. 13. Fitting situation.

Table 4
Degree of Fit.

	Coefficient of Determination (R^2)	Root Mean Square
w	0.9911	3.4253
c_1	0.8583	0.0788

Table 5
Changes in boundary conditions under strong time-varying conditions.

Location	Parameter	Operating conditions 1	Operating conditions 2	Operating conditions 3
The gas side of AHE1	Temperature/K	723	-	723
	Mass flow rate/(kg/s)	0.48	-	0.48
The gas side of AHE2	Temperature/K	633	603	633
	Mass flow rate /(kg/s)	0.37	0.32	0.37

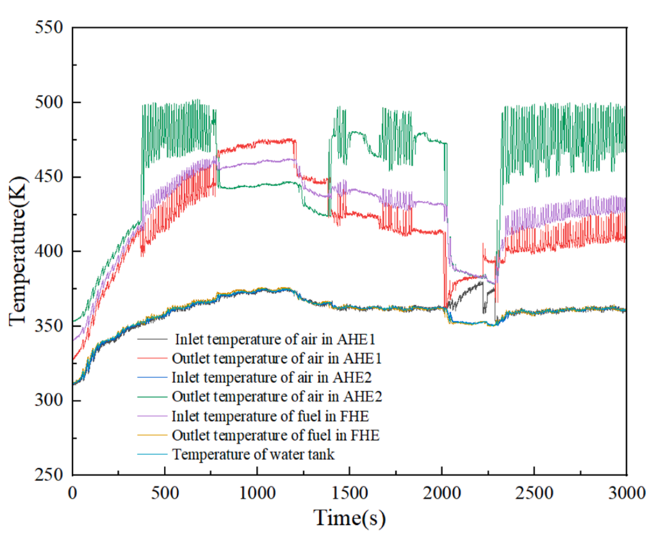


Fig. 14. Water temperature at each point of the test stand during strong time-varying experiments.

Table 6
Working conditions for different time periods.

Working conditions	1	2	3
Time period	0–2000s	2000–2300s	2300–3000

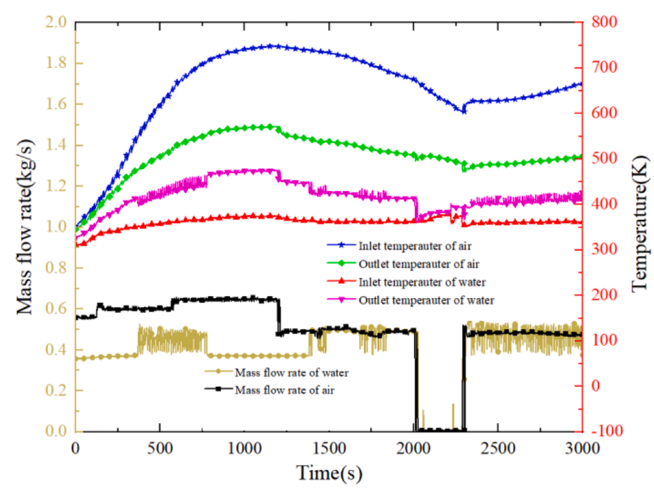


Fig. 15. Inlet and outlet temperatures and flow rates on both sides of AHE1.

corresponding to the left axis, while the inlet and outlet temperatures of the fluids on both sides are shown above the legends, with the temperatures corresponding to the right axis. Several phenomena can be observed: (1) The temperature of the air is difficult to stabilize and keeps changing, especially during the first 800 s; (2) At around 2000 s and 2300 s, a working condition switch was carried out once; (3) During the time periods of 350 s – 750 s, 1300 s – 1700 s and 2300 s – 3000 s, the temperature or flow rate fluctuations were relatively severe.

As shown in Fig. 18, this is the experimental result of the strong time-varying experiment at 1000–3000 s. In the figure, the first and second rows represent the flow conditions and inlet temperatures of the air side and fuel side of the three heat exchangers respectively. From this, it is clearly seen that the changes in boundary conditions. The third row shows the control results, and the fourth row shows the changes in control variables. Thus, the effect of control can be seen. At 2000 s and 2300 s, it is the moments of two working condition changes. Due to the requirement that the intermediate circulating working medium does not

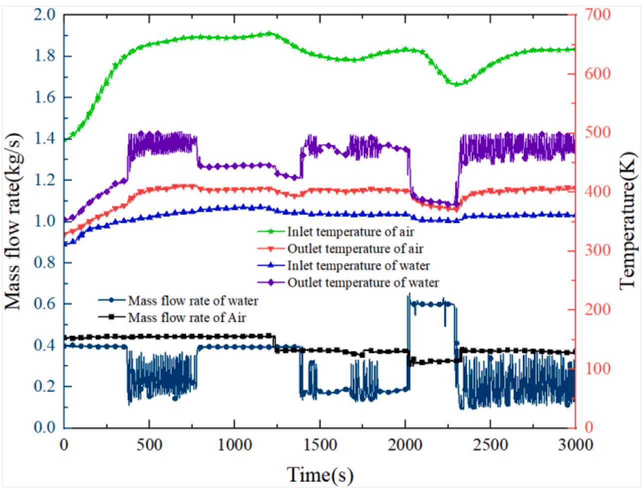


Fig. 16. Inlet and outlet temperatures and flow rates on both sides of AHE2.

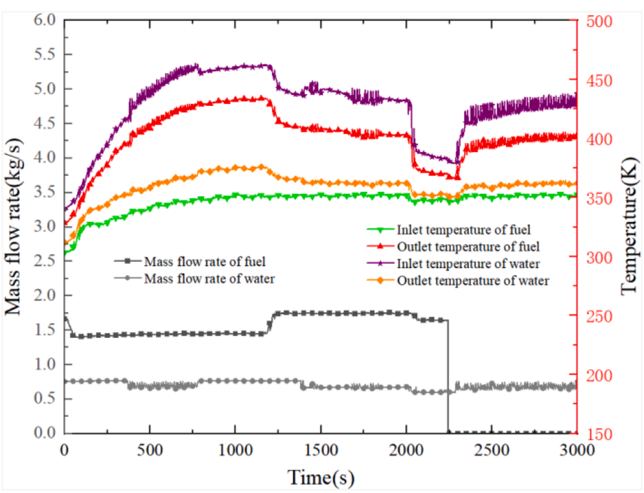


Fig. 17. Inlet and outlet temperatures and flow rates on both sides of FHE.

boil, it can be seen from the line graph of the valve opening change in the bottommost layer that the deviation between the algorithm control and the actual control is large. Only at 1500–2000 s is the control effect better. At this time, the working medium water has not reached the boiling temperature (seen in Fig. 14, in which the highest water temperature reaches about 500 K, and the critical boiling point).

For the time period from 1500 to 2000 s, the actual outlet temperature was compared with the target temperature. Quantitative results are presented in Table 7. During this interval, the MPC controller achieve effective control performance in practice, with a maximum deviation of 3.43 K, a time-averaged absolute deviation of 0.96 K and a root mean square deviation of 1.2 K. The maximum deviation occurs when the valve opening calculated by the controller differs from the actual operating value. At this moment (1773.6 s), the system has already entered the rule-based protection logic designed to prevent water boiling. As can be observed in Fig. 18, a discrepancy between the calculated and actual valve positions leads to fluctuations in the air-side outlet temperature, resulting in a relatively large deviation.

Furthermore, the experimental results demonstrate the efficacy of the hybrid control architecture, where the rule-based anti-boiling logic overrides the MPC when temperatures approach the critical limit. This switching mechanism is crucial for absolute constraint satisfaction, as the supervisor takes pre-validated, conservative actions to guarantee safety. Although a formal hybrid-system stability analysis is beyond the scope of this study, practical stability is confirmed empirically: the overrides are transient and minimally intrusive, with the system consistently returning to MPC operation once safe conditions are restored, all while maintaining the overall control objective of outlet temperature regulation, as evidenced by the performance metrics in Table 7.

6. Conclusion

This paper conducts heat current modeling for the intermediate loop in the FTMS of aircraft engines with time delays, and designs a constrained MPC approach based on the heat current model. The heat transfer constraints of nonlinearity are considered completely by using the heat current method. For the intermediate circulating loop of FTMS of aircraft engines containing water tank, pipeline, and heat exchanger, the flow delay of the fluid in the pipeline is considered, and the effects of temperature and flow rate changes on the heat transfer at the non-fuel side of the heater are taken into account, so that the heat current models of the fuel tank, the pipeline, and the simplified dynamic heat exchanger are established respectively, and the heat current model of the whole aircraft engine FTMS is completed based on the topological relationship between the systems.

Based on the heat current model, the state space expression of the system is further constructed and discretized by the forward Euler method. On this basis, considering the delay characteristics of the system, the construction of the discrete generalized state space model is completed by expanding the state matrix in the time domain and introducing the control error feedback. Subsequently, the cost function containing constraints is designed and transformed into a QP problem for solving, and the first step control quantity is extracted from the optimal control sequence and applied to the system, realizing the design of the MPC controller with constraints.

Furthermore, a full-scale ground-test experimental system has been constructed to replicate real flight conditions. Using this platform, the key characteristic parameters for the control algorithm were identified from test data. These include the flow regulation characteristics of pump-valve assemblies, heat transfer performance of heat exchangers, dynamic capacity of the water tank, and system delay characteristics. Based on these identified parameters, a complete MPC controller was constructed and integrated into the experimental system.

Finally, after simulating the strong time-varying flight state and realizing the automatic control test, the algorithm was verified to be practically controllable. Under the premise of ensuring that the intermediate circulatory system work water does not boil during the whole experiment, the temperature requirements of the air outlet on the air side are met as much as possible. Quantitative results show the controller maintains air outlet temperature with a maximum absolute deviation of 3.43 K, a time-averaged absolute deviation of 0.96 K, and a root mean square deviation of 1.20 K, while effectively preventing the intermediate circulating water from boiling throughout the experiments.

In summary, the principal strength of this work is the experimental validation of a control framework that successfully handles coupled nonlinear and time-delayed dynamics in real-time. The practical demonstration under strong time-varying conditions confirms the framework's capability to manage complex thermal-fluid interactions that are challenging for conventional methods.

Furthermore, the model's key assumptions—including the constant delay time, the uniformly mixed water tank, and the use of an effective thermal resistance with delay potentials for heat exchangers—warrant further discussion regarding their collective impact. The constant delay assumption primarily induces a phase prediction error during rapid flow variations, while the simplified HX model may lead to inaccuracies in capturing high-frequency thermal transients. However, in our MPC design, these unmodeled dynamics are effectively compensated by the closed-loop feedback correction and the empirically tuned parameters, which prioritize robustness over aggressive performance. This is evidenced by the robust control performance achieved under strong time-varying conditions (see Section 4.2, Table 7). Regarding the tank

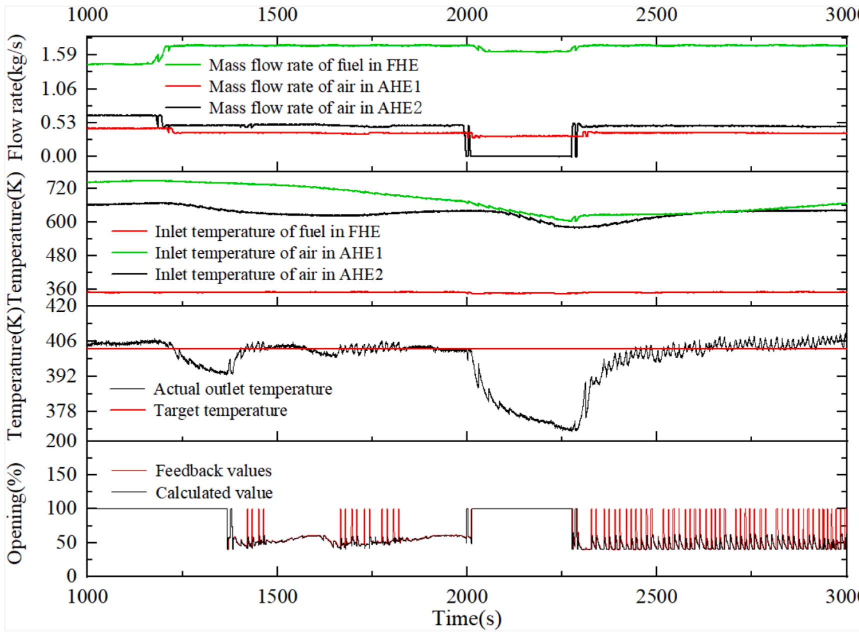


Fig. 18. Strong time-varying experimental result plots.

Table 7
The quantitative situation of control performance during the 1500s to 2000s.

Maximum absolute deviation /K	Time-averaged absolute deviation /K	Root mean square/K
3.43	0.96	1.20

assumption, the identified dynamic capacity (~11 kg) and operational flow rates (e.g., 0.37–0.65 kg/s) imply a fluid residence time between approximately 17–30 s. Since the critical thermal fluctuations and control actions occur over longer periods (e.g., >100 s), the uniform temperature assumption remains valid for capturing the dominant system dynamics. The experimental results, showing a maximum output error of 3.43 K and an average error of 0.96 K, serve as a practical bound on the overall error induced by all model simplifications, confirming their acceptability for the control objectives of this study.

Despite the demonstrated performance, this study has certain limitations that point to future research directions. Firstly, the fluid flow delay time was assumed to be constant, independent of the mass flow rate. While this simplification facilitated the derivation of a linearized predictive model for MPC, it may introduce prediction errors during periods of highly transient flow. Future work will incorporate a flow-dependent delay model to enhance accuracy. Secondly, the heat exchanger model, while capturing essential nonlinearities and delays, is a simplified representation of the full transient dynamics. A more detailed dynamic model could be explored to capture higher-frequency thermal behaviors, albeit at an increased computational cost.

Notwithstanding these limitations, this research fundamentally advances the field of thermal management control. It establishes a novel framework that integrates the heat current method with MPC, providing a systematic and foundational solution for handling coupled nonlinear heat transfer and time-delay effects. The component-based methodology offers an extensible modeling tool for diverse system architectures. This work thus establishes a critical benchmark and provides a validated technical foundation for the development of next-generation control strategies in aerospace and other complex energy systems.

CRediT authorship contribution statement

Ke-Lun He: Writing – review & editing, Writing – original draft, Methodology. **Tian-Yi Zhang:** Data curation. **Xiao-Guang Zhang:** Investigation. **Qun Chen:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for this journal and was not involved in the editorial review or the decision to publish this article.

Acknowledgements

The present work is supported by the National Natural Science Foundation of China (Grant No. 52306016, 52125604 and 52250273).

Appendix A. The expressions for the 36 partial derivatives of the elements in Eqs. (18), (20), (21), (22) and (24)

The expressions for the 36 partial derivatives of the elements in Eqs. (18), (20), (21), (22) and (24) is given by:

$$\frac{\partial \ddot{m}_0}{\partial T_\mathrm{T}} = \frac{\partial \ddot{m}_0}{\partial \dot{m}_0} = \frac{\partial \ddot{m}_0}{\partial \dot{m}_1} = \frac{\partial \ddot{m}_1}{\partial T_\mathrm{T}} = \frac{\partial \ddot{m}_1}{\partial \dot{m}_0} = \frac{\partial \ddot{m}_1}{\partial \dot{m}_1} = 0 \tag{A.1}$$

$$\frac{\partial \dot{T}_\mathrm{T}}{\partial \dot{m}_0(t-\tau_2)} = \frac{\partial \dot{T}_\mathrm{T}}{\partial \dot{m}_1(t-\tau_2)} = \frac{\partial \ddot{m}_0}{\partial T_\mathrm{T}(t-\tau_2)} = \frac{\partial \ddot{m}_0}{\partial \dot{m}_0(t-\tau_2)} = 0 \tag{A.2}$$

$$\frac{\partial \ddot{m}_0}{\partial \dot{m}_1(t-\tau_2)}=\frac{\partial \ddot{m}_1}{\partial T_{\mathrm{T}}(t-\tau_2)}=\frac{\partial \ddot{m}_1}{\partial \dot{m}_0(t-\tau_2)}=\frac{\partial \ddot{m}_1}{\partial \dot{m}_1(t-\tau_2)}=0 \tag{A.3}$$

$$\frac{\partial \dot{T}_{\mathrm{T}}}{\partial \ddot{m}_0}=\frac{\partial \dot{T}_{\mathrm{T}}}{\partial \ddot{m}_1}=\frac{\partial \ddot{m}_0}{\partial \ddot{m}_1}=\frac{\partial \ddot{m}_1}{\partial \ddot{m}_0}=\frac{\partial T_{\mathrm{hout1}}}{\partial T_{\mathrm{T}}}=\frac{\partial T_{\mathrm{hout2}}}{\partial T_{\mathrm{T}}}=\frac{\partial T_{\mathrm{hout1}}}{\partial \ddot{m}_0}=0 \tag{A.4}$$

$$\frac{\partial \ddot{m}_0}{\partial \ddot{m}_0}=\frac{\partial \ddot{m}_1}{\partial \ddot{m}_1}=1 \tag{A.5}$$

$$\frac{\partial T_{\mathrm{hout1}}}{\partial \dot{m}_0(t-\tau_{\mathrm{b}})}=\frac{\partial T_{\mathrm{hout1}}}{\partial \dot{m}_1(t-\tau_{\mathrm{b}})}=\frac{\partial T_{\mathrm{hout2}}}{\partial \dot{m}_0(t-\tau_{\mathrm{b}})}=\frac{\partial T_{\mathrm{hout2}}}{\partial \dot{m}_1(t-\tau_{\mathrm{b}})}=0 \tag{A.6}$$

$$\frac{\partial \dot{T}_{\mathrm{T}}}{\partial T_{\mathrm{T}}}=-\frac{\dot{m}_0}{m_{\mathrm{T}}} \tag{A.7}$$

$$\frac{\partial T_{\mathrm{hout1}}}{\partial \dot{m}_0}=\frac{\left[e^{a_{\mathrm{h1}}}-(1-a_{\mathrm{c1}})e^{a_{\mathrm{c1}}}\right](G_{\mathrm{c1}}e^{a_{\mathrm{h1}}}-G_{\mathrm{h1}}e^{a_{\mathrm{c1}}})-\dot{m}_1\left(c_{\mathrm{water}}e^{a_{h1}}+\frac{G_{\mathrm{h1}}a_{\mathrm{c1}}}{\dot{m}_1}e^{a_{\mathrm{c1}}}\right)(e^{a_{\mathrm{h1}}}-e^{a_{\mathrm{c1}}})}{(G_{\mathrm{c1}}e^{a_{\mathrm{h1}}}-G_{\mathrm{h1}}e^{a_{\mathrm{c1}}})^2}*c_{\mathrm{water}}[T_{\mathrm{T}}(t-\tau_a)-T_{\mathrm{hin1}}(t-\tau_{\mathrm{H1}})] \tag{A.8}$$

$$\frac{\partial T_{\mathrm{hout1}}}{\partial T_{\mathrm{T}}(t-\tau_{\mathrm{b}})}=\frac{G_{\mathrm{c1}}(e^{a_{\mathrm{h1}}}-e^{a_{\mathrm{c1}}})}{G_{\mathrm{c1}}e^{a_{\mathrm{h1}}}-G_{\mathrm{h1}}e^{a_{\mathrm{c1}}}} \tag{A.9}$$

$$\frac{\partial T_{\mathrm{hout2}}}{\partial \dot{m}_0}=\frac{(e^{a_{\mathrm{h2}}}-e^{a_{\mathrm{c2}}}+a_{\mathrm{c2}}e^{a_{\mathrm{c2}}})(G_{\mathrm{c2}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})-(\dot{m}_0-\dot{m}_1)(\dot{m}_0c_{\mathrm{water}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})(e^{a_{\mathrm{h2}}}-e^{a_{\mathrm{c2}}})}{(G_{\mathrm{c2}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})^2}*c_{\mathrm{water}}[T_{\mathrm{T}}(t-\tau_{\mathrm{b}})-T_{\mathrm{hin2}}(t-\tau_{\mathrm{H2}})] \tag{A.10}$$

$$\frac{\partial T_{\mathrm{hout2}}}{\partial \dot{m}_1}=-\frac{(e^{a_{\mathrm{h2}}}-e^{a_{\mathrm{c2}}}+a_{\mathrm{c2}}e^{a_{\mathrm{c2}}})(G_{\mathrm{c2}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})-(\dot{m}_0-\dot{m}_1)(\dot{m}_0c_{\mathrm{water}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})(e^{a_{\mathrm{h2}}}-e^{a_{\mathrm{c2}}})}{(G_{\mathrm{c2}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})^2}*c_{\mathrm{water}}[T_{\mathrm{T}}(t-\tau_{\mathrm{b}})-T_{\mathrm{hin2}}(t-\tau_{\mathrm{H2}})] \tag{A.11}$$

$$\frac{\partial T_{\mathrm{hout2}}}{\partial T_{\mathrm{T}}(t-\tau_{\mathrm{b}})}=\frac{G_{\mathrm{c2}}(e^{a_{\mathrm{h2}}}-e^{a_{\mathrm{c2}}})}{G_{\mathrm{c2}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}}} \tag{A.12}$$

$$\begin{aligned}
\frac{\partial \dot{T}_{\mathrm{T}}}{\partial \dot{m}_0}=&\frac{G_{\mathrm{h2}}(T_{\mathrm{hin2}}-T_{\mathrm{T}}(t-\tau_2))\left[(e^{a_{\mathrm{h2}}}-e^{a_{\mathrm{c2}}}+a_{\mathrm{c2}}e^{a_{\mathrm{c2}}})(G_{\mathrm{c2}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})-(\dot{m}_0-\dot{m}_1)(e^{a_{\mathrm{h2}}}-e^{a_{\mathrm{c2}}})\left(e^{a_{\mathrm{h2}}}c_{\mathrm{water}}+\frac{G_{\mathrm{h2}}a_{\mathrm{c2}}e^{a_{\mathrm{c2}}}}{\dot{m}_0-\dot{m}_1}\right)\right]}{m_{\mathrm{T}}(G_{\mathrm{c2}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})^2}+\\
&\frac{T_{\mathrm{T}}(t-\tau_2)}{m_{\mathrm{T}}}-\frac{T_{\mathrm{T}}}{m_{\mathrm{T}}}+\frac{\dot{m}_1G_{\mathrm{h1}}G_{\mathrm{c3}}(e^{a_{\mathrm{h1}}}-e^{a_{\mathrm{c1}}})(T_{\mathrm{hin1}}-T_{\mathrm{T}}(t-\tau_2))\left[-\frac{a_{\mathrm{h3}}}{\dot{m}_0}e^{a_{\mathrm{h3}}}(G_{\mathrm{c3}}e^{a_{\mathrm{h3}}}-G_{\mathrm{h3}}e^{a_{\mathrm{c3}}})+(e^{a_{\mathrm{h3}}}-e^{a_{\mathrm{c3}}})\left(\frac{G_{\mathrm{c3}}a_{\mathrm{h3}}}{\dot{m}_0}e^{a_{\mathrm{h3}}}+c_{\mathrm{water}}e^{a_{\mathrm{c3}}}\right)\right]}{m_{\mathrm{T}}(G_{\mathrm{c1}}e^{a_{\mathrm{h1}}}-G_{\mathrm{h1}}e^{a_{\mathrm{c1}}})(G_{\mathrm{c3}}e^{a_{\mathrm{h3}}}-G_{\mathrm{h3}}e^{a_{\mathrm{c3}}})^2}+\\
&\frac{G_{\mathrm{c3}}G_{\mathrm{h2}}(T_{\mathrm{hin2}}-T_{\mathrm{T}}(t-\tau_2))(G_{\mathrm{c3}}e^{a_{\mathrm{h3}}}-G_{\mathrm{h3}}e^{a_{\mathrm{c3}}})(\dot{m}_0c_{\mathrm{water}}e^{a_{\mathrm{h2}}}-\dot{m}_1c_{\mathrm{water}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})}{m_{\mathrm{T}}[(G_{\mathrm{c3}}e^{a_{\mathrm{h3}}}-G_{\mathrm{h3}}e^{a_{\mathrm{c3}}})(\dot{m}_0c_{\mathrm{water}}e^{a_{\mathrm{h2}}}-\dot{m}_1c_{\mathrm{water}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})]^2}*\\
&\left[\left(1-\frac{a_{\mathrm{h3}}}{\dot{m}_0}+\frac{\dot{m}_1a_{\mathrm{h3}}}{\dot{m}_0}\right)e^{a_{\mathrm{h2}}}e^{a_{\mathrm{h3}}}+\left(a_{\mathrm{h3}}+a_{\mathrm{c2}}-1-\frac{\dot{m}_1a_{\mathrm{h3}}}{\dot{m}_0}\right)e^{a_{\mathrm{c2}}}e^{a_{\mathrm{h3}}}-e^{a_{\mathrm{h2}}}e^{a_{\mathrm{c3}}}+(1-a_{\mathrm{c2}})e^{a_{\mathrm{c2}}}e^{a_{\mathrm{c3}}}\right]-\\
&\frac{G_{\mathrm{c3}}G_{\mathrm{h2}}(T_{\mathrm{hin2}}-T_{\mathrm{T}}(t-\tau_2))(\dot{m}_0-\dot{m}_1)(e^{a_{\mathrm{h3}}}-e^{a_{\mathrm{c3}}})(e^{a_{\mathrm{h2}}}-e^{a_{\mathrm{c2}}})}{m_{\mathrm{T}}[(G_{\mathrm{c3}}e^{a_{\mathrm{h3}}}-G_{\mathrm{h3}}e^{a_{\mathrm{c3}}})(\dot{m}_0c_{\mathrm{water}}e^{a_{\mathrm{h2}}}-\dot{m}_1c_{\mathrm{water}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})]^2}*\\
&\left[\left(1-a_{\mathrm{h3}}+\frac{\dot{m}_1a_{\mathrm{h3}}}{\dot{m}_0}\right)G_{\mathrm{c3}}c_{\mathrm{water}}e^{a_{\mathrm{h2}}}e^{a_{\mathrm{h3}}}+\left(\frac{a_{\mathrm{h3}}}{\dot{m}_0}+\frac{a_{\mathrm{c2}}}{\dot{m}_0-\dot{m}_1}\right)G_{\mathrm{c3}}G_{\mathrm{h2}}e^{a_{\mathrm{c2}}}e^{a_{\mathrm{h3}}}+(\dot{m}_1-2\dot{m}_0)c_{\mathrm{water}}^2e^{a_{\mathrm{h2}}}e^{a_{\mathrm{c3}}}+\left(1-\frac{\dot{m}_0a_{\mathrm{c2}}}{\dot{m}_0-\dot{m}_1}\right)e^{a_{\mathrm{c2}}}e^{a_{\mathrm{c3}}}\right]-\\
&\frac{G_{\mathrm{c3}}(T_{\mathrm{T}}(t-\tau_2)-T_{\mathrm{cin3}})\left[(e^{a_{\mathrm{h3}}}-a_{\mathrm{h3}}e^{a_{\mathrm{h3}}}-e^{a_{\mathrm{c3}}})(G_{\mathrm{c3}}e^{a_{\mathrm{h3}}}-G_{\mathrm{h3}}e^{a_{\mathrm{c3}}})+(\dot{m}_0e^{a_{\mathrm{h3}}}-\dot{m}_0e^{a_{\mathrm{c3}}})\left(\frac{G_{\mathrm{c3}}a_{\mathrm{h3}}}{\dot{m}_0}e^{a_{\mathrm{h3}}}+c_{\mathrm{water}}e^{a_{\mathrm{c3}}}\right)\right]}{m_{\mathrm{T}}(G_{\mathrm{c3}}e^{a_{\mathrm{h3}}}-G_{\mathrm{h3}}e^{a_{\mathrm{c3}}})^2}
\end{aligned} \tag{A.13}$$

$$\begin{aligned}
\frac{\partial \dot{T}_{\mathrm{T}}}{\partial \dot{m}_1}=&\frac{G_{\mathrm{h1}}(T_{\mathrm{hin1}}-T_{\mathrm{T}}(t-\tau_2))\left[(e^{a_{\mathrm{h1}}}-e^{a_{\mathrm{c1}}}+a_{\mathrm{c1}}e^{a_{\mathrm{c1}}})(G_{\mathrm{c1}}e^{a_{\mathrm{h1}}}-G_{\mathrm{h1}}e^{a_{\mathrm{c1}}})-\dot{m}_1(e^{a_{\mathrm{h1}}}-e^{a_{\mathrm{c1}}})\left(c_{\mathrm{water}}e^{a_{\mathrm{h1}}}+\frac{a_{\mathrm{c1}}G_{\mathrm{h1}}}{\dot{m}_1}e^{a_{\mathrm{c1}}}\right)\right]}{m_{\mathrm{T}}(G_{\mathrm{c1}}e^{a_{\mathrm{h1}}}-G_{\mathrm{h1}}e^{a_{\mathrm{c1}}})^2}-\\
&\frac{G_{\mathrm{h2}}(T_{\mathrm{hin2}}-T_{\mathrm{T}}(t-\tau_2))\left[(e^{a_{\mathrm{h2}}}-e^{a_{\mathrm{c2}}}+a_{\mathrm{c2}}e^{a_{\mathrm{c2}}})(G_{\mathrm{c2}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})-(\dot{m}_0-\dot{m}_1)(e^{a_{\mathrm{h2}}}-e^{a_{\mathrm{c2}}})\left(e^{a_{\mathrm{h2}}}c_{\mathrm{water}}+\frac{G_{\mathrm{h2}}a_{\mathrm{c2}}e^{a_{\mathrm{c2}}}}{\dot{m}_0-\dot{m}_1}\right)\right]}{m_{\mathrm{T}}(G_{\mathrm{c2}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})^2}-\\
&\frac{G_{\mathrm{h1}}G_{\mathrm{c3}}(e^{a_{\mathrm{h3}}}-e^{a_{\mathrm{c3}}})(T_{\mathrm{hin1}}-T_{\mathrm{T}}(t-\tau_2))\left[(e^{a_{\mathrm{h1}}}-e^{a_{\mathrm{c1}}}+a_{\mathrm{c1}}e^{a_{\mathrm{c1}}})(G_{\mathrm{c1}}e^{a_{\mathrm{h1}}}-G_{\mathrm{h1}}e^{a_{\mathrm{c1}}})-\dot{m}_1(e^{a_{\mathrm{h1}}}-e^{a_{\mathrm{c1}}})\left(c_{\mathrm{water}}e^{a_{\mathrm{h1}}}+\frac{G_{\mathrm{h1}}a_{\mathrm{c1}}}{\dot{m}_1}e^{a_{\mathrm{c1}}}\right)\right]}{m_{\mathrm{T}}(G_{\mathrm{c3}}e^{a_{\mathrm{h3}}}-G_{\mathrm{h3}}e^{a_{\mathrm{c3}}})(G_{\mathrm{c1}}e^{a_{\mathrm{h1}}}-G_{\mathrm{h1}}e^{a_{\mathrm{c1}}})^2}-\\
&\frac{G_{\mathrm{c3}}(e^{a_{\mathrm{h3}}}-e^{a_{\mathrm{c3}}})G_{\mathrm{h2}}(T_{\mathrm{hin2}}-T_{\mathrm{T}}(t-\tau_2))\left[(e^{a_{\mathrm{c2}}}-e^{a_{\mathrm{h2}}}-a_{\mathrm{c2}}e^{a_{\mathrm{c2}}})(G_{\mathrm{c2}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})+(\dot{m}_0-\dot{m}_1)(e^{a_{\mathrm{h2}}}-e^{a_{\mathrm{c2}}})\left(c_{\mathrm{water}}e^{a_{\mathrm{h2}}}+\frac{G_{\mathrm{h2}}ac2}{\dot{m}_0-\dot{m}_1}e^{a_{\mathrm{c2}}}\right)\right]}{m_{\mathrm{T}}(G_{\mathrm{c3}}e^{a_{\mathrm{h3}}}-G_{\mathrm{h3}}e^{a_{\mathrm{c3}}})(G_{\mathrm{c2}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})^2}
\end{aligned} \tag{A.14}$$

$$\frac{\partial \dot{T}_{\mathrm{T}}}{\partial T_{\mathrm{T}}(t-\tau_2)} = -\frac{\dot{m}_1 G_{\mathrm{h1}}(e^{a_{\mathrm{h1}}}-e^{a_{\mathrm{c1}}})}{m_{\mathrm{T}}(G_{\mathrm{c1}}e^{a_{\mathrm{h1}}}-G_{\mathrm{h1}}e^{a_{\mathrm{c1}}})} - \frac{(\dot{m}_0-\dot{m}_1)G_{\mathrm{h2}}(e^{a_{\mathrm{h2}}}-e^{a_{\mathrm{c2}}})}{m_{\mathrm{T}}(G_{\mathrm{c2}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})} + \frac{\dot{m}_0}{m_{\mathrm{T}}} + \frac{\dot{m}_1 G_{\mathrm{c3}}(e^{a_{\mathrm{h3}}}-e^{a_{\mathrm{c3}}})G_{\mathrm{h1}}(e^{a_{\mathrm{h1}}}-e^{a_{\mathrm{c1}}})}{m_{\mathrm{T}}(G_{\mathrm{c3}}e^{a_{\mathrm{h3}}}-G_{\mathrm{h3}}e^{a_{\mathrm{c3}}})(G_{\mathrm{c1}}e^{a_{\mathrm{h1}}}-G_{\mathrm{h1}}e^{a_{\mathrm{c1}}})}$$
$$+\frac{(\dot{m}_0-\dot{m}_1)G_{\mathrm{c3}}(e^{a_{\mathrm{h3}}}-e^{a_{\mathrm{c3}}})G_{\mathrm{h2}}(e^{a_{\mathrm{h2}}}-e^{a_{\mathrm{c2}}})}{m_{\mathrm{T}}(G_{\mathrm{c3}}e^{a_{\mathrm{h3}}}-G_{\mathrm{h3}}e^{a_{\mathrm{c3}}})(G_{\mathrm{c2}}e^{a_{\mathrm{h2}}}-G_{\mathrm{h2}}e^{a_{\mathrm{c2}}})} - \frac{\dot{m}_0 G_{\mathrm{c3}}(e^{a_{\mathrm{h3}}}-e^{a_{\mathrm{c3}}})}{m_{\mathrm{T}}(G_{\mathrm{c3}}e^{a_{\mathrm{h3}}}-G_{\mathrm{h3}}e^{a_{\mathrm{c3}}})} \tag{A.15}$$

Data availability

Data will be made available on request.

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